

Machines acquire scientific taste from institutional traces

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Abstract

Artificial intelligence matches or exceeds human performance on tasks with verifiable answers, from protein folding to Olympiad mathematics. Yet the capacity that most governs scientific advance is not reasoning but taste: the ability to judge which untested ideas deserve pursuit, exercised daily by editors and funders but never successfully articulated, taught, or automated. Here we show that fine-tuning language models on journal publication decisions recovers evaluative judgment inaccessible to both frontier models and human expertise. Using a held-out benchmark of research pitches in management spanning four quality tiers, we find that eleven frontier models, spanning major proprietary and open architectures, average 31% accuracy yet discriminate among tiers worse than random guessing. Panels of journal editors and editorial board members reach 42% by majority vote. Fine-tuned models trained on years of publication records each surpass every frontier model and expert panel, with the best single model achieving 59%. These models exhibit calibrated confidence, reaching 100% accuracy on their highest-confidence predictions, and transfer this evaluative signal to untrained pairwise comparisons and one-sentence summaries. The mechanism generalizes: models trained on economics publication records achieve 70% accuracy. Scientific taste was not missing from AI’s reach; it was deposited in the institutional record, waiting to be extracted. These results provide a scalable mechanism to triage the expanding volume of scientific production across disciplines where quality resists formal verification.

Introduction

Science is built on verifiable facts, but it is governed by unverifiable judgments. Artificial intelligence now matches or exceeds human performance on tasks with objectively verifiable answers, from protein structure prediction[1] to Olympiad-level mathematics[2] to competitive programming[3]. Yet the scientific enterprise depends equally on a form of taste, not reasoning: the ability to discriminate which untested ideas are worth pursuing. This scientific taste is tacit[8], socially constituted[20], and transmitted through exposure to what the field considers excellent, not through explicit rules. As AI systems generate hypotheses and manuscripts at unprecedented scale[4, 5], the bottleneck in science has shifted from production to evaluation[6]. Whether AI can acquire this form of taste (subjective, contextual, resistant to explicit codification) defines the next critical boundary in machine intelligence, where capability drops sharply from domains with clear answers to those requiring discrimination and judgment[7].

Evaluative judgment is a paradigmatic instance of tacit knowledge: “we can know more than we can tell”[8]. Decades of effort to codify what makes a research idea good (structured review rubrics, editorial frameworks distinguishing originality from utility[9, 10]) have not produced agreement. Reviewers agree on categorical assignment at barely above chance[11]; the gap between what the system achieves collectively and what any individual reviewer contributes is vast. Neither formal training, academic rank[13], nor editorial experience[14] reliably predicts review quality; review performance actually declines with experience[15]. Ethnographic work reveals that academic judgment operates through intuitive assessment and disciplinary sensibility, not rule application[16]. Novel ideas, precisely the ones that matter most, generate the greatest disagreement[17], and the system systematically undervalues its most impactful work[18].

The quality signal is real: it manifests in the long-run sorting of publications across prestige tiers[19]. That signal is nonetheless irreducibly tacit, not in Polanyi’s original sense of individual embodied skill,

but in what Collins terms *collective* tacit knowledge[20]: evaluative understanding embedded in institutional practices that no individual participant can articulate, yet that the system reliably enacts[8, 21]. In Bourdieu’s terms, this is the *habitus* of a scientific field, a disposition acquired through prolonged immersion in what the field rewards, not through rules[48]. Explicit evaluative criteria do not improve inter-rater reliability; the knowledge resists codification even when codification is attempted[11]. Yet the institutional system nonetheless produces consistent quality stratification over time. This prolonged operation has left a trace: years of publication decisions across prestige tiers constitute an institutional record of accumulated scientific taste — a form of ‘dark knowledge’[38] (evaluative information implicit in institutional outcomes but absent from any explicit criterion) that has never been exploited as a training signal for artificial intelligence.

This tacit character exposes a deep asymmetry in artificial intelligence. In domains with verifiable outputs, AI systems achieve superhuman performance[1, 2, 3]; but this advantage vanishes when the task shifts from verification to evaluation. When deployed in peer review, a setting where more than half of researchers already report using AI[22, 23], large language models consistently inflate scores, recommend acceptance at rates far exceeding human panels, and systematically overlook novelty, the dimension that most distinguishes important from incremental work[24, 25, 26]. They evaluate how research is presented, not what it contributes[27]. This leniency is not a prompting failure but a predictable consequence of preference-based post-training: reinforcement learning (RL) from human preferences instills sycophantic behavior that intensifies with further optimization[28, 29], producing fluent models that default to approval instead of the discrimination that taste demands.

Here we construct a held-out benchmark of 120 article-derived research pitches in organizational psychology and management¹, balanced across four quality tiers and evaluated in 2,914 human ratings from 48 expert gatekeepers and 174 junior researchers, and replicate the approach in economics with a 200-article benchmark. We test 26 AI configurations spanning frontier reasoning models, chat models, supervised fine-tuned models, architecture-matched base controls, and reinforcement-learning ablations, generating over 18,900 independent evaluation events under an expert-derived prompt optimized to favor frontier performance. We also evaluate auxiliary pairwise discrimination tasks to test whether this evaluative capacity generalizes beyond the training format. Frontier models perform only marginally above chance (31.1% versus 25%). Rather than discriminating across quality levels, most collapse their predictions into one or two categories — six of eleven never identify a single article as low quality, defaulting instead to generically favourable assessments. Human evaluators perform somewhat better and preserve more balanced tier discrimination, but their judgments remain highly variable, with inter-rater agreement near zero. Neither prompting the most powerful AI systems nor assembling expert panels solves the evaluation problem.

We align AI to these *institutional traces*. Individual reviewers disagree substantially, yet the institutional system, operating through iterated consensus over time, produces reliable quality stratification that individual judgments do not[19]. It is this system-level signal, not individual reviewer accuracy, that supervised learning can exploit[20]. By applying supervised fine-tuning (SFT) to four base models using historical research-pitch/journal-outcome pairs, we align them to implicit evaluative criteria that neither written instructions nor the largest AI architectures have successfully transmitted.

The results are clear. All four fine-tuned models independently achieve 55.0–59.2% accuracy, and a simple two-model pair extends this to 60.8%, capturing approximately six times the headroom between

¹ Organizational psychology and management is a large, globally distributed field (over 20,000 Academy of Management members; 1,800+ ranked journals in the Chartered ABS Academic Journal Guide). Top-tier journals maintain acceptance rates of 4–10%, creating a well-documented institutional hierarchy. Research quality is evaluated primarily on theoretical contribution and conceptual novelty, not resource access, isolating judgments of idea quality from confounds such as laboratory infrastructure. Meta-analytic inter-rater reliability is low ($\kappa = 0.17$ across 48 studies[11]), consistent with evaluation that is genuinely tacit.

chance and ceiling recovered by frontier models and more than double that of the expert majority vote, at a total training cost below \$300. Cross-architecture replication confirms that the effect is not model-specific: each single model exceeds both the frontier mean and the best frontier model. Beyond category prediction, the fine-tuned models exhibit calibrated self-knowledge: accuracy exceeds 80% in the top-confidence sixth and reaches 100% in the top 10% of predictions, giving the benchmark's strongest practically useful signal for distinguishing correct from incorrect judgments. Without training on pairwise discrimination, the fine-tuned models transfer evaluative judgment to head-to-head pitch comparisons, a task format not present in training. The larger fine-tuned model further retains this evaluative signal when full idea summaries are compressed to one-sentence inputs not encountered during training, still exceeding the best frontier model and expert panels and demonstrating learned evaluative understanding instead of memorized tier assignments. When trained on economics institutional traces, the same approach yields 69.5% accuracy, confirming that the mechanism generalizes beyond the primary field.

These findings carry implications for the acceleration of science. Scientific taste — the capacity to judge which questions are worth asking — is the binding constraint on scientific discovery, most acutely in the social and behavioral sciences, where the value of research questions resists formal verification and submission volumes far outstrip expert bandwidth. Cross-field replication in economics confirms that the mechanism is not specific to management. These are precisely the fields where frontier models fail most completely and where scalable, calibrated evaluation would yield the greatest returns. The mechanism should generalize to other domains with socially judged outcomes and weak ex ante verification, including creative industries, venture investing, grant allocation, hiring and promotion, and policy execution, whenever institutional decision histories accumulate large records of what later proved successful[19, 32]. In our benchmark, frontier-model performance clusters close to chance across the full breadth of proprietary and open-weight architectures, yet a fine-tuned system trained on accumulated editorial decisions exceeds both the most powerful AI systems and the expert panels whose collective decisions generated the training data. The barrier to AI evaluation is not raw capability alone; it is the absence of the right training signal.

Results

A benchmark for evaluative judgment

Peer review rests on a cognitive capacity that has resisted formal specification: the ability to make early-stage significance judgments before full empirical execution, when journals, funders, and labs decide where to invest scarce attention and resources. To isolate this capacity, we constructed a held-out benchmark in management comprising 120 article-derived research pitches, balanced across four quality tiers and spanning 15 research domains and 17 journals (of the 19-journal source universe; see Methods and Supplementary Methods SM5). The source articles were all published after June 30, 2025. Each source article was transformed into a research pitch centered on the core research question and theoretical framing, with detailed methods, full empirical findings, and publication identifiers removed. Articles were assigned to four quality tiers based on their publication outlet (exceptional, strong, fair, limited; 30 per tier; chance baseline 25%; Fig. 1). Because evaluators see only the research question and theoretical framing (not methods, results, or journal identity), tier assignment requires the kind of discrimination closest to taste: judging an idea's promise from its presentation alone.

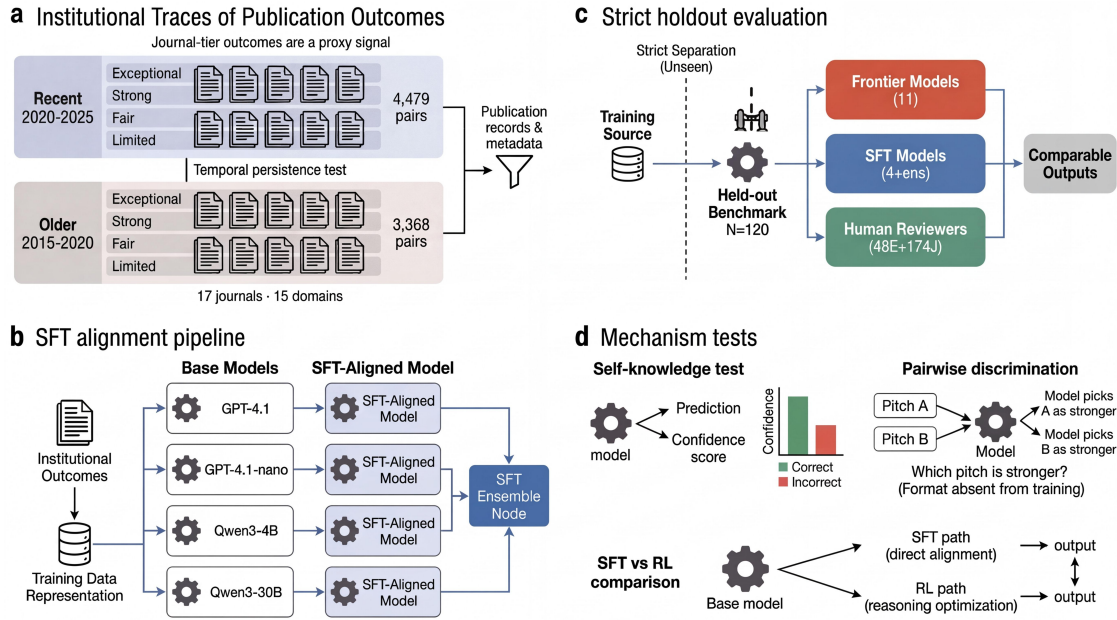


Figure 1 | Study design: institutional traces, alignment pipeline, evaluation, and mechanism tests

a, Institutional traces of publication outcomes. Journal-tier publication outcomes (exceptional, strong, fair, limited) serve as the supervision signal for supervised fine-tuning (SFT). Two temporally disjoint training-data slices are constructed from a 19-journal source universe spanning 15 research domains: a recent slice (2020–2025; 4,479 research-pitch/journal-outcome pairs) used for the primary SFT models and an older slice (2015–2020; 3,368 pairs) used to test the temporal persistence of the institutional signal. Both slices are fully disjoint from the 120-pitch held-out benchmark. **b**, SFT alignment pipeline. Institutional outcomes are converted into standardized training representations and used to fine-tune four base models spanning two model families and multiple parameter scales (GPT-4.1, GPT-4.1-nano, Qwen3-4B, Qwen3-30B). Each base model produces an architecture-matched SFT checkpoint; checkpoint outputs are combined via probability averaging into an SFT ensemble node. **c**, Strict holdout evaluation. Training data and the held-out benchmark ($N = 120$ pitches, 30 per tier) are strictly separated. Three evaluator classes are assessed on the same unseen benchmark under a common four-tier output mapping: 11 frontier reasoning models, 4 SFT models plus ensembles (ens.), and human reviewers comprising 48 expert gatekeepers (E) and 174 junior researchers (J). All evaluators use the same frozen expert-derived prompt and identical tier definitions, ensuring that performance differences reflect evaluator capability rather than task-format variation. **d**, Mechanism tests. Three auxiliary analyses probe how SFT acquires evaluative judgment. *Self-knowledge test*: each model produces a tier prediction together with a calibrated confidence score; the test asks whether confidence is systematically higher on correct predictions than on incorrect ones. *Pairwise discrimination*: two research pitches are presented side by side and the model identifies the stronger one, a task format entirely absent from training, testing whether SFT develops generalized quality ordering that transfers beyond the categorical classification used during fine-tuning. *SFT versus reinforcement learning (RL) comparison*: models from the same family are trained via direct supervised alignment (SFT) or reasoning-optimized RL (using reasoning-enabled configurations), and outputs are compared to test whether explicit chain-of-thought deliberation aids or impedes evaluative accuracy on tacit judgment tasks.

Evaluators were asked to assign each pitch to one of the four tiers using an expert-derived assessment framework centered on originality and utility[9, 10]. The full evaluator pool comprised eleven frontier reasoning models, three chat variants, four supervised fine-tuned (SFT) models with architecture-matched base controls, 48 expert gatekeepers, and 174 junior researchers, generating 2,914 human ratings and over 16,000 AI inferences across 26 model configurations (including prompt-sensitivity, pairwise discrimination, and reinforcement-learning ablations), for a combined total exceeding 18,900 independent evaluation events. The evaluation prompt was selected as the formulation that maximized frontier model performance, ensuring that any advantage observed for fine-tuned models represents a conservative estimate.

Frontier models and the anatomy of prediction collapse

If evaluative judgment were reducible to the kind of reasoning that frontier language models excel at (synthesizing information, recognizing patterns in text, applying criteria to cases), then the most capable

models available should perform well above chance on this benchmark, particularly when given maximal advantage. We gave them that advantage: an evaluation prompt incorporating expert-derived assessment criteria[9, 10], stress-tested across variants to maximize frontier performance (Extended Data Fig. 1).

It is not enough. Across a cohort of 11 frontier models, including Gemini 3.1 Pro, Claude Opus 4.6, GPT-5.2 High, Gemini 2.5 Pro, Qwen 3.5 Plus, Grok 4.1 Fast, and leading open-weight systems such as Kimi K2.5, DeepSeek V3.2 and GLM-5, mean accuracy was 31.1%, barely exceeding the 25% chance baseline and capturing only 8.1% of the available headroom between chance and perfect accuracy² (Fig. 2a; see Methods for statistical tests). The failure runs deeper than accuracy alone: mean macro F1, a measure of balanced classification quality that penalizes strategies ignoring minority tiers, was 0.236, below the 0.25 expected from random guessing. Models achieve above-chance accuracy on the tiers they over-predict while producing zero recall on tiers they ignore, a strategy that inflates accuracy but destroys balanced discrimination.



Figure 2 | Frontier models show structured prediction collapse

² Gemini 3.1 Pro reached the highest frontier accuracy (38.8%) but its response occasionally reproduced benchmark content instead of returning a tier label, raising benchmark-leakage concerns; we retain it with explicit caveating instead of treating it as a clean comparator. To provide a cleaner within-family reference, the frontier cohort also includes Gemini 2.5 Pro, released in June 2025. Even so, no frontier model reliably separates from the cohort after multiple-testing correction.

a, Eleven frontier models are compared with paired accuracy and macro-F1 bars under the conservative primary protocol, with Gemini 3.1 retained but explicitly marked for contamination risk. Accuracy bars show 95% binomial confidence intervals ($n = 120$ per model), while macro-F1 shows bootstrap 95% confidence intervals from per-model predictions. The main point is that the full frontier cohort remains close to chance despite using the best-performing prompt formulation. **b**, Predicted-tier distributions (100% stacked) show heavy concentration in the strong and fair tiers and sparse use of the limited tier, making the collapse pattern visible beyond scalar accuracy. **c**, Row-normalized confusion matrices for four representative flagships show distinct failure modes: Gemini 3.1 as the best-performing but still compressed model, GPT-5.2 High with middle-tier clustering, Claude Opus 4.6 with strong-tier ceiling behavior, and Qwen 3.5 Plus as an additional reference model showing that collapse is not confined to the three highlighted proprietary flagships. Extended collapse diagnostics across all 11 models are shown in Supplementary Fig. 4.

The failure follows three distinct patterns of prediction collapse that reveal how post-training alignment shapes model behavior on evaluative tasks (Fig. 2b). The first pattern is strong-ceiling collapse: Claude Opus 4.6 assigns 87% of articles to the “strong” tier, producing a macro F1 of just 0.145, predicting virtually everything as above-average quality, using only two of the four available categories. The second is leniency-to-middle collapse: Seed 2.0 Pro and Grok 4.1 heavily over-assign above-average tiers. The third is middle-tier clustering: GPT-5.2 High and DeepSeek V3.2 concentrate almost all predictions in the strong and fair tiers, compressing the distribution into a narrow band. Across the cohort, 6 of 11 models never predict “limited” for any article, producing zero recall on the lowest quality tier (Fig. 2c; full confusion matrices for all 11 models in Extended Data Fig. 3a; class-level distribution diagnostics in Supplementary Figs. 1, 4). These patterns are consistent with reinforcement learning from human feedback optimizing outputs toward agreeable, broadly acceptable responses[28, 29], a property that serves conversational fluency but systematically undermines the discriminating judgments that scientific gatekeeping demands. The evaluation prompt represented the most thorough attempt to transmit human evaluative criteria through explicit instruction. Its failure across all 11 models suggests that scientific taste resists propositional transmission: it cannot be told, only shown.

Human expertise and the limits of domain knowledge

The frontier models’ failure might reflect a limitation specific to artificial systems. Perhaps evaluative judgment requires the kind of tacit knowledge that only years of immersion in a research community can provide[8]. To test this, we recruited 48 expert gatekeepers³: current editors and editorial board members at leading journals in organizational behavior and management, individuals whose professional role consists precisely in making the quality judgments this benchmark demands.

Expert reviewers did outperform frontier models, but not by the margin their credentials would predict. Individual expert accuracy averaged 36.2%, significantly above chance but reflecting enormous variability: some experts performed well below chance while others in one case reached 100% on their rated articles (Fig. 3a; Supplementary Fig. 2; see Methods for tests of significance). When expert ratings were aggregated through majority vote, accuracy rose to 41.6% on the 89 articles that produced a clear plurality (31 of 120 articles yielded tied votes and were excluded), capturing 22.1% of the available headroom (Fig. 3b). Experts achieved macro F1 values of 0.353 (individual average) and 0.404 (majority vote), while the 11-model frontier average remained 0.236 (below the 0.25 random baseline), demonstrating that humans genuinely discriminate across quality tiers instead of collapsing predictions into a narrow band (Supplementary Fig. 6).

³ Each of the 120 benchmark pitches was rated by a mean of 3.2 experts, matching the typical reviewer load for a journal submission (2–3 independent reviewers per manuscript). The 48-rater panel thus provides per-pitch evaluation depth comparable to real editorial decisions while covering the full benchmark.

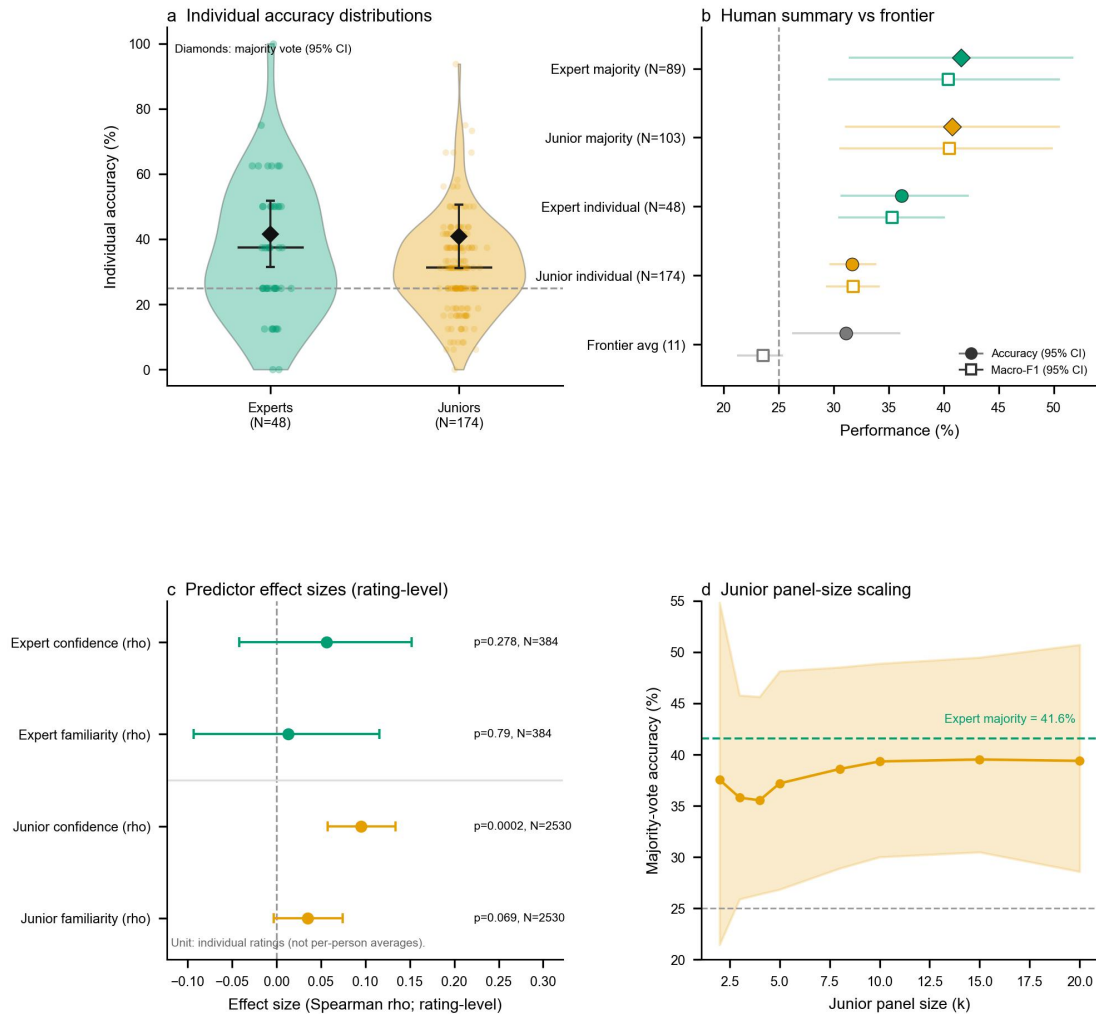


Figure 3 | Human expertise helps, but noise remains dominant

a, Individual expert and junior accuracy distributions are shown with majority-vote reference markers, highlighting that experts outperform juniors on average but both groups remain highly heterogeneous; Supplementary Fig. 2 and Supplementary Table ST9 provide the full expert distribution details. **b**, Human reference configurations are compared against the frontier-model baseline on accuracy and macro-F1 with 95% uncertainty intervals, showing that human raters preserve more balanced discrimination than the frontier cohort. **c**, Confidence and topic familiarity are summarized as rating-level Spearman effect estimates on correctness, with bootstrap confidence intervals and permutation-based P values; the panel makes clear that these self-reported expertise markers explain little of the observed variation in accuracy. **d**, Monte Carlo matched-N junior subsampling shows how majority-vote accuracy changes with panel size, revealing early gains followed by a plateau rather than unlimited improvement; Supplementary Fig. 3 and Supplementary Table ST10 provide the corresponding resampling details.

The pattern of agreement among experts exposes a notable dissociation. Categorical agreement, whether experts assign articles to the same tier, was near-chance (Fleiss' kappa = 0.047), barely distinguishable from noise[11]. Yet ordinal agreement, whether experts rank articles in roughly the same order of quality, was moderate (Krippendorff's alpha = 0.307). This gap suggests that experts share a coarse sense of which articles are better or worse but disagree substantially on where to draw categorical boundaries, consistent with evidence that expert judgment is dominated by noise that persists regardless of training or experience[30].

We searched systematically for expertise markers that might predict which reviewers perform well. No expertise marker predicted accuracy: not career stage, self-reported confidence, nor topic familiarity

(Fig. 3c). This pattern matches a broader literature showing that neither domain expertise[33], career stage, nor publication record[13] reliably predicts evaluation accuracy.

Junior researchers (174 doctoral and postdoctoral scholars) performed comparably. Individual accuracy averaged 31.7%, and full majority vote reached 40.8% on the 103 articles with a clear plurality. Subsampling analysis, repeatedly drawing expert-sized panels from the junior researcher pool, yielded accuracy not significantly different from expert majority vote (Fig. 3d; Supplementary Fig. 3). Panel-size analysis revealed that accuracy plateaus at approximately 10 reviewers per article (Extended Data Fig. 4), indicating a structural ceiling on what aggregation alone can achieve.

These results establish that evaluative judgment is genuinely difficult, not merely for AI systems optimized for conversational agreeableness, but for the human experts whose professional authority rests on this very capacity. The quality signal exists: institutional publication outcomes confirm it. But neither the most capable reasoning systems in artificial intelligence nor decades of accumulated domain expertise can reliably recover that signal from research pitches alone. A different kind of learning might succeed: one that acquires taste from the institutional record instead of reasoning over explicit criteria.

Fine-tuning on institutional traces recovers scientific taste

We hypothesized that the information needed to discriminate quality tiers is not absent from language models' latent representations but rather inaccessible under standard prompting and reasoning regimes. To test this, we applied supervised fine-tuning (SFT) to four base models using 4,479 historical research-pitch/journal-outcome pairs from the primary recent/new institutional-trace slice, the data that encode the field's accumulated gatekeeping consensus[19]. The 120 benchmark research pitches were held out entirely and not seen during training. SFT does not inject new factual knowledge; it aligns latent representations to the distributional regularities through which institutional quality manifests in research pitches (see Methods for training details).

Before fine-tuning, we evaluated each base model on the benchmark to establish architecture-matched controls. Three of four base models scored at or near chance level (22.5–26.7%); GPT-4.1 reached 32.5%, modestly above chance but far below its fine-tuned counterpart (Extended Data Table 1). These results confirm that architecture alone cannot extract evaluative signal: the smaller Qwen3-4B scored near chance before fine-tuning yet improved by 32.5 percentage points after, while Qwen3-30B-A3B showed the largest gain in the study at 35.8 percentage points (Extended Data Fig. 3b).

Fine-tuning produced a marked improvement. All four models achieved 55.0–59.2% accuracy, gains of 22.5 to 35.8 percentage points over their respective base models (Fig. 4a). That these gains emerge consistently across two model families and two parameter scales confirms that the effect is not an artifact of a particular architecture, scale, or training implementation.

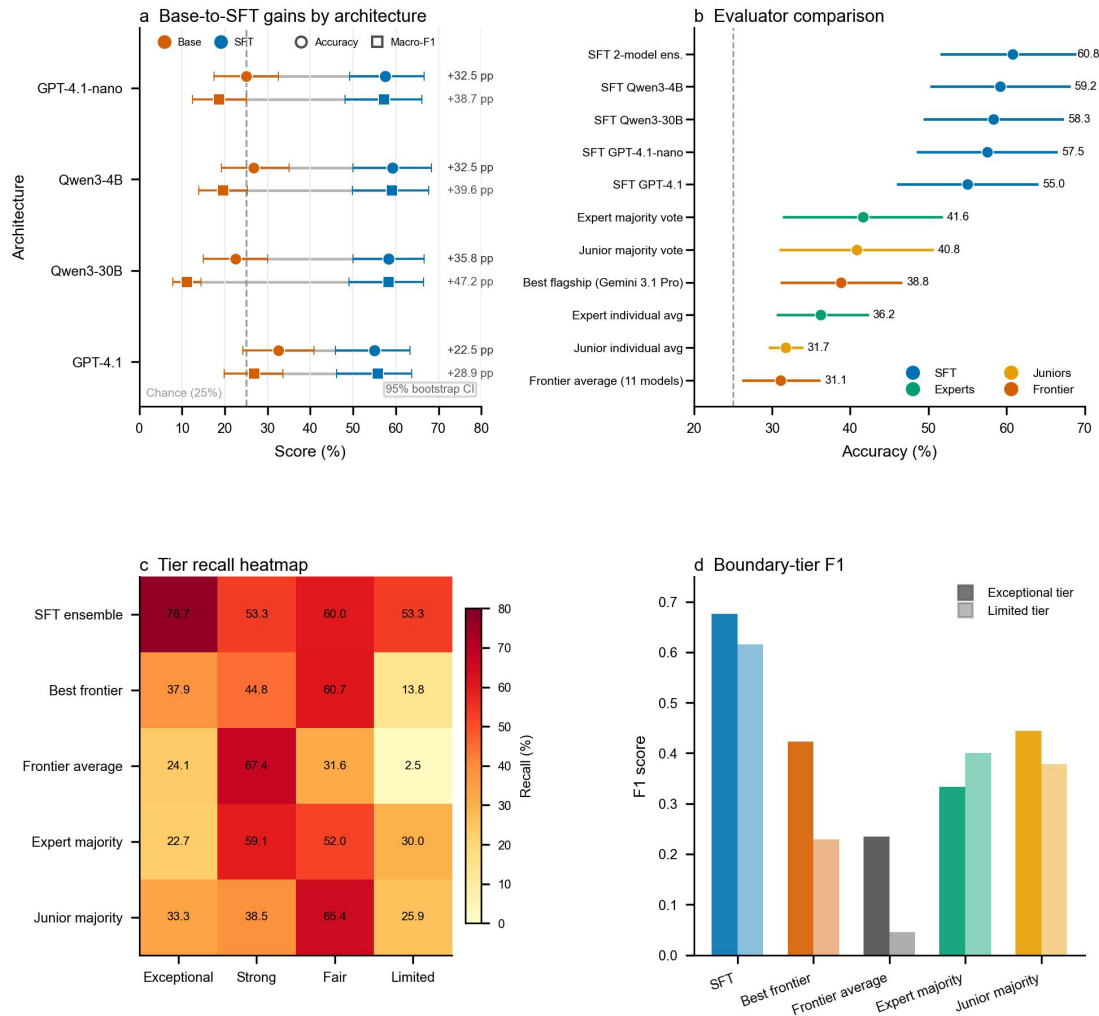


Figure 4 | SFT on institutional traces recovers tier discrimination

a, Four architecture-matched base-to-SFT pairs (GPT-4.1, GPT-4.1-nano, Qwen3-30B, Qwen3-4B) are shown as linked points, with circles for accuracy and squares for macro-F1. Error bars are bootstrap 95% confidence intervals recomputed from the model prediction files used in this analysis, and the annotated percentage-point gains isolate the effect of supervised fine-tuning from architecture choice alone. **b**, An accuracy-only evaluator comparison places the SFT 2-model ensemble, all four single SFT checkpoints, expert and junior individual/majority references, the best frontier model, and the frontier average on the same axis with 95% confidence intervals from the benchmark summary tables. **c**, Row-normalized recall across the four quality tiers compares the SFT ensemble, best frontier model, frontier average, expert majority vote, and junior majority vote, showing that SFT recovers the full tier structure rather than collapsing toward the middle. **d**, Boundary-tier F1 for the exceptional and limited tiers highlights where SFT gains are strongest and why they matter for gatekeeping decisions at the two extremes of article quality.

The selected primary pair, GPT-4.1-nano (SFT) and Qwen3-30B-A3B (SFT), achieved 60.8% (95% CI: 51.7–69.2%), exceeding the best single Qwen3-4B checkpoint by 1.7 percentage points and capturing 47.8% of the headroom between chance and ceiling. The frontier average captures 8.1%; expert majority vote captures 22.1%. The SFT ensemble therefore captures approximately six times the headroom of frontier models and more than double that of expert panels, with a macro F1 of 0.607 reflecting balanced discrimination across all four tiers (Supplementary Table ST3). We systematically evaluated all six pairwise ensemble combinations among the four SFT checkpoints via probability averaging; accuracies ranged from 59.2% to 60.8%, and all six combinations exceeded the frontier average (31.1%) by 28.1 to 29.8 percentage points (Supplementary Table ST4; Extended Data Fig. 3c).

Comparisons confirmed the robustness of these differences. Against the frontier average (31.1%), the SFT ensemble outperformed by +29.8 percentage points (exact binomial, $p = 1.74 \times 10^{-11}$); against the best-performing frontier model under conservative protocol (Gemini 3.1 Pro, 39.1% majority accuracy on paired subset), by +20.9 percentage points (McNemar test, $p = 0.001425$; Supplementary Table ST8). Because no individual frontier model significantly separates from the cohort after multiple-testing correction, the comparison against the frontier mean remains the most representative. Against individual evaluators, the ensemble surpassed the typical expert by a large margin (60.8% vs. 36.2% mean, $t(47) = -8.32$, $p < 0.001$, $d = 1.20$), with 83% of experts and 97% of junior researchers scoring below the ensemble (Fig. 4b). Against collective majority votes, SFT showed advantages of 21.4 percentage points over expert majority ($p = 0.00729$) and 21.4 percentage points over junior majority ($p = 0.00196$).

The per-tier analysis reveals where the advantage concentrates and exposes a fundamental distinction in evaluation style (Fig. 4c,d). SFT achieved higher balanced detection rates (F1) than expert panels at all four tiers, with the largest gain at the exceptional tier (0.68 versus 0.33) and continued improvements at the limited (0.62 versus 0.40), strong (0.58 versus 0.46), and fair (0.55 versus 0.43) tiers. Human panels remain conservative gatekeepers: high precision when they commit to an extreme judgment (63% for exceptional, 60% for limited), but missing the majority of articles at each quality end (23% recall for exceptional, 30% for limited). The SFT ensemble predicts much closer to the benchmark base rate (38 exceptional, 25 strong, 35 fair, 22 limited), and its false positives at the exceptional tier remain mostly near-misses from the adjacent strong tier (10 of 15), not catastrophic misclassifications. In any system where the cost of missing exceptional work exceeds the cost of additional review, the typical condition when submission volumes far outstrip expert bandwidth, this broader coverage addresses the more consequential failure mode.

To test whether this signal reflects durable evaluative structure or transient patterns, we trained matched architectures on an older institutional trace slice (2015–2020, 3,368 pairs), introducing a five-year lag relative to the benchmark. The Qwen3-30B SFT trained on the older slice reached 46.7%, still exceeding both the frontier average (31.1%) and expert majority vote (41.6%), versus 58.3% for its recent-slice counterpart (Extended Data Fig. 6). What drifts is the quality bar, not the signal: the older model over-predicted exceptional articles by 50%, reflecting competitive thresholds that have since risen as submission volumes grew and acceptance rates fell [44, 45]. The model learned the standards of its era faithfully; those standards have since been raised.

Total training cost across all four models was under \$300, spanning two cloud API fine-tuning jobs (~\$200 and ~\$10 for GPT-4.1 and GPT-4.1-nano, no hardware required) and two local GPU runs (~1 and ~8 A100 hours for Qwen3-4B and Qwen3-30B-A3B; Supplementary Table ST2). The critical ingredient behind the main benchmark gains is the training signal, institutional traces encoding accumulated scientific taste, not model scale alone. Model scale matters mainly for how robustly that learned signal generalizes under severe input compression.

The mechanism: training signal drives performance; model scale shapes generalization

The SFT results pose a mechanistic puzzle. Why does a simple training procedure on journal-tier labels unlock evaluative performance that neither the largest frontier models nor experienced human reviewers achieve? Four lines of evidence clarify the mechanism and its boundary conditions: direct example-based alignment to institutional traces drives the benchmark gains across architectures, whereas model scale mainly governs how robustly the learned representation generalizes beyond the full idea-summary format.

Calibrated metacognition. A defining property of genuine expertise is knowing what one knows, the capacity to assign higher confidence to judgments that prove correct. The SFT ensemble exhibited this property: its prediction confidence was significantly higher on correct predictions than on incorrect ones (confidence gap = +0.082, $p = 0.0081$; expected calibration error = 0.073; Fig. 5a). All four individual

SFT models showed the same pattern, although the Qwen3-4B gap was much smaller, confirming that calibration is a consistent property of the fine-tuning approach, not an artifact of a single model.

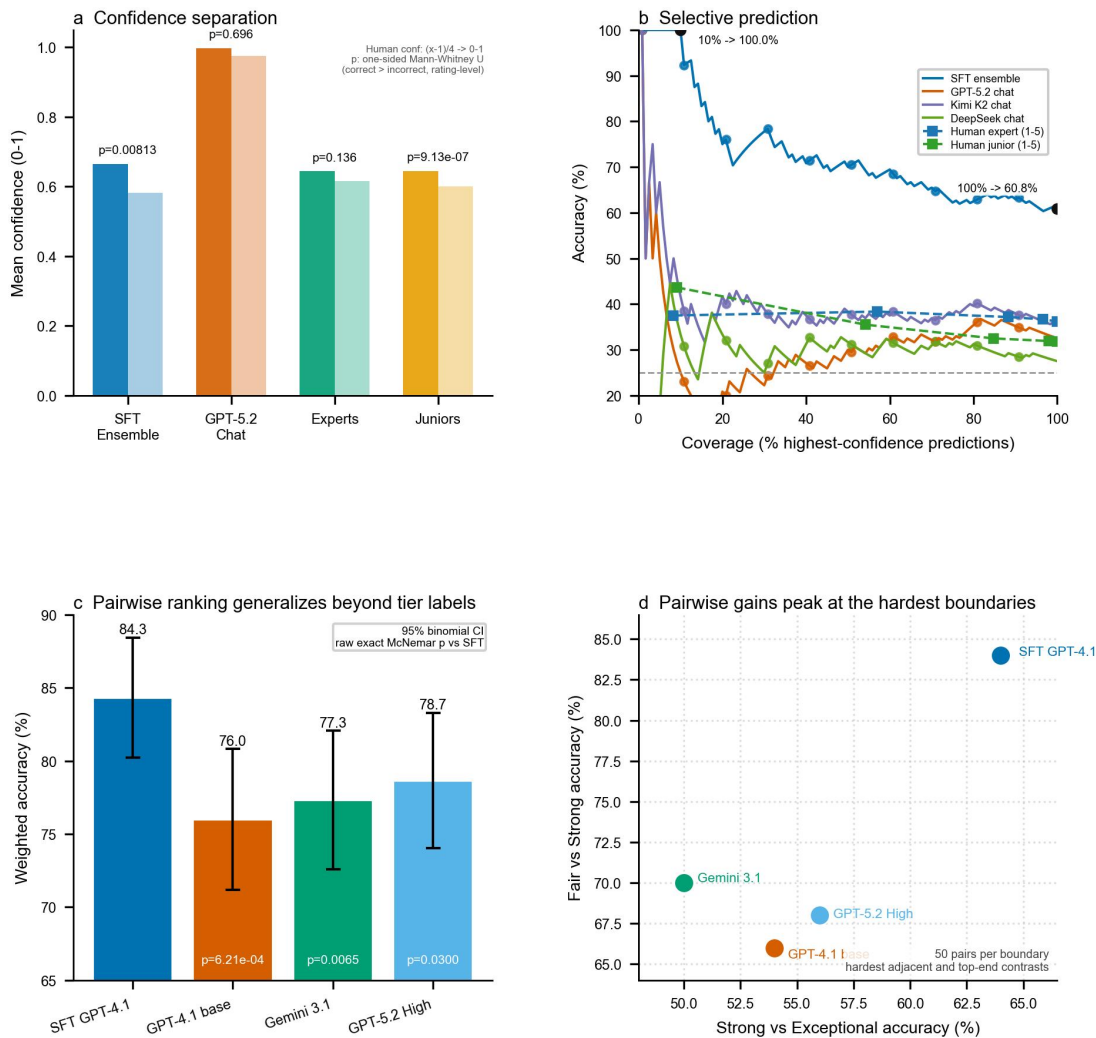


Figure 5 | SFT mechanism: self-knowledge and generalized pairwise ranking

a, Mean confidence for correct and incorrect predictions is compared for the SFT ensemble, GPT-5.2 chat, experts, and juniors. Human 1-5 confidence is mapped to 0-1 by $(x - 1) / 4$, and P values are one-sided Mann-Whitney U tests comparing whether correct ratings are more confident than incorrect ratings at the rating level. The panel asks whether each evaluator knows when it is right. **b**, Selective-prediction curves show how accuracy changes as lower-confidence cases are deferred, with AI predictions ordered from highest to lowest confidence and human curves defined by confidence thresholds; this panel visualizes the practical triage value of calibrated abstention. **c**, Overall weighted accuracy on the fixed 300-pair pairwise evaluation set compares the shared four-model subset: SFT GPT-4.1, Gemini 3.1 Pro, GPT-5.2 High, and the GPT-4.1 baseline, with 95% binomial confidence intervals and raw unadjusted exact paired McNemar P values versus SFT. **d**, A hard-boundary summary isolates the two most informative pair types, strong_exceptional and fair_strong, showing that the SFT pairwise advantage is concentrated where relative ranking is hardest. Extended Data Fig. 2 retains the six-pair-type heatmap and discordant-pair decomposition for the same plotted subset.

This calibration stands in sharp contrast to both frontier models and human evaluators (Fig. 5a; Extended Data Fig. 4a-c). GPT-5.2 chat exhibited extreme, undifferentiated overconfidence: near-maximum confidence regardless of whether its predictions were correct, despite near-chance accuracy (ECE = 0.657, a measure of how well stated confidence matches actual accuracy; Extended Data Fig. 4b). Human expert confidence was similarly non-discriminative. Junior researcher confidence showed a statistically detectable but practically weak rating-level association with accuracy. Frontier models and expert raters

therefore provide no comparably useful self-screening signal, and the detectable junior effect remains too weak to support high-precision abstention.

The SFT models can. When predictions were restricted to the top sixth by confidence (16.7% coverage), accuracy reached 80.0%; at the top 10.0% (12 of 120 articles), accuracy reached 100%; every high-confidence prediction was correct (Fig. 5b). This selective prediction capacity, the ability to flag cases where the model’s judgment is highly reliable while abstaining on uncertain ones, represents the strongest practically useful metacognitive triage signal in the benchmark. In practice, this enables a confidence-based workflow: route high-confidence predictions directly and flag uncertain cases for human review.

Pairwise head-to-head discrimination. To confirm that SFT develops generalized evaluative judgment, not category-specific pattern matching, we tested models in a pairwise format entirely absent from training. Models received two research pitches simultaneously and judged which represented the stronger research pitch. The main figure pairwise panels focus on the shared four-model subset used throughout the pairwise analysis: SFT GPT-4.1, Gemini 3.1 Pro, GPT-5.2 High, and the GPT-4.1 baseline. Within this shared subset, SFT GPT-4.1 reached 84.3% (253/300), versus 77.3% (232/300) for Gemini 3.1 Pro, 78.7% (236/300) for GPT-5.2 High, and 76.0% (228/300) for the GPT-4.1 baseline (Fig. 5c; Supplementary Table ST1). Fine-tuning on institutional traces thus exceeded both the model’s own pre-training baseline and the frontier comparators in this auxiliary pairwise set. The gains were not uniform across pair types: the largest margins appeared at the fair-strong and strong-exceptional boundaries, where SFT reached 84.0% and 64.0% versus 70.0% and 50.0% for Gemini 3.1 Pro, 68.0% and 56.0% for GPT-5.2 High, and 66.0% and 54.0% for the GPT-4.1 baseline (Fig. 5d). Extended Data Fig. 2 provides the full heatmap across all six pair types and the discordance decomposition for the same plotted subset. On the same 300 paired items, raw unadjusted exact McNemar tests showed significant SFT advantages versus Gemini 3.1 Pro ($p = 0.00646$), GPT-5.2 High ($p = 0.0300$), and GPT-4.1 baseline ($p = 0.000621$; Fig. 5c).

SFT versus reinforcement learning. If evaluative judgment is tacit and resists decomposition into reasoning steps[8], training methods that optimize articulated reasoning chains may perform worse than methods that learn directly from examples. Cognitive science supports this prediction: verbal articulation of holistic judgments degrades their quality[35], and chain-of-thought prompting yields negligible or negative gains outside mathematics and symbolic reasoning[36, 37]. We tested this by training reinforcement learning variants on the same base architectures (see Methods). RL run-pooled accuracy was 40.3%, above the frontier average (31.1%) but below every SFT single model (55.0–59.2%; Extended Data Fig. 5). When the model’s reasoning chain diverged from its final label, accuracy dropped markedly; when reasoning and label agreed, performance approached SFT levels. Explicit deliberation can override otherwise sound evaluative intuitions[35, 49]. SFT learns taste through exposure to exemplars, not through reasoning about criteria.

Transfer under input compression. A natural objection to the SFT results is that the models may have learned superficial patterns in the rich contextual descriptions used during training, not genuine evaluative judgment. We tested this by evaluating the same fine-tuned checkpoints on an input format completely different from the one used in training: one-sentence idea statements stripped of theoretical framing, methodological detail, and contribution claims (for examples of paired one-sentence and full-summary inputs, see Supplementary Methods SM4). The larger GPT-4.1 fine-tuned model retained substantial evaluative signal under this compression, achieving 49.2% accuracy on one-sentence inputs versus 55.0% on the full summary (Extended Data Fig. 7a). This compressed input accuracy exceeds the best frontier model on the full input (Gemini 3.1 Pro, 38.8%) and the expert majority vote (41.6%), demonstrating that the evaluative representation learned through fine-tuning survives radical reduction of the input signal. The transfer was not uniform: recall at the exceptional and strong tiers declined while

recall at the limited tier rose from 60.0% to 83.3%, indicating a conservative shift under uncertainty, not random degradation (Extended Data Fig. 7b,c).

The smaller GPT-4.1-nano fine-tuned model showed a sharply different pattern. Despite matching GPT-4.1 SFT on the full input (57.5% versus 55.0%), it collapsed to 33.3% under compression, barely above chance and indistinguishable from its own base model (30.8%). This asymmetry indicates that acquiring the evaluative signal and generalizing it across input formats are separable capabilities: the same institutional training signal improves both models on the full benchmark, but greater representational capacity makes that learned signal more robust under severe compression.

Together, these lines of evidence (calibrated self-knowledge, generalized pairwise discrimination, transfer under input compression, and the SFT advantage over RL) converge on a more specific conclusion: supervised fine-tuning on institutional traces extracts a genuine evaluative representation. The training signal drives the benchmark gains across architectures, while model scale shapes how robustly the learned representation transfers under severe input compression.

Voting consistency and aggregation asymmetry

If individual SFT models capture evaluative signal, a natural question is whether aggregating across models amplifies that signal further, and whether the same holds for frontier models and human panels. The final mechanism question is whether aggregation improves judgment through additional votes alone, or through the quality and diversity of the voters. Fig. 6 presents a unified comparison of filtering by consensus across AI and human evaluator classes, revealing a fundamental asymmetry in how aggregation functions across evaluator types.

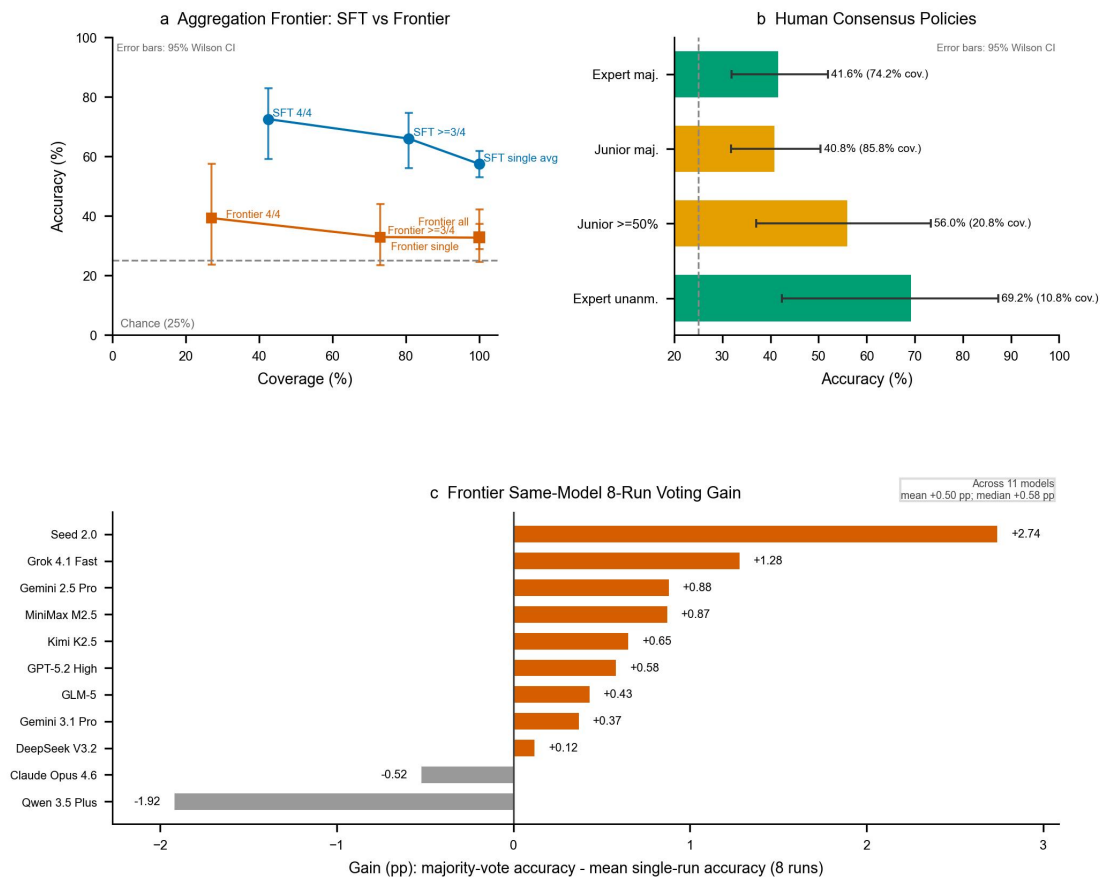


Figure 6 | Aggregation asymmetry after metric-definition alignment

a, Cross-model consensus policies are compared for the four SFT models and a fixed four-model frontier set. Accuracy intervals use Wilson 95% confidence intervals, and coverage indicates the share of articles satisfying each agreement rule. The contrast is substantive: stricter consensus improves SFT precision sharply, whereas frontier cross-model voting remains weak even at higher agreement thresholds. **b**, Human consensus policies compare expert majority, junior majority, junior $\geq 50\%$ vote share, and expert unanimity among articles with at least two expert ratings. Accuracy intervals in this panel also use Wilson 95% confidence intervals, and labels report the associated coverage, making the human accuracy-coverage tradeoff explicit. **c**, Same-model 8-run gains for frontier systems are summarized as majority-vote accuracy minus mean single-run accuracy, showing that repeated sampling alone yields only modest improvements and does not solve the underlying judgment problem.

Across the four SFT models, full consensus (4/4) occurs on 42.5% of articles ($N = 51$) and reaches 72.5% accuracy, the highest precision achieved by any evaluator configuration in this study (Fig. 6a). Within this unanimous subset, accuracy is sharpest at the quality extremes: 100.0% for the exceptional tier and 66.7% for the limited tier, while articles in the strong tier remain hardest (53.8%; fair 66.7%; Supplementary Table ST12). Relaxing to $\geq 3/4$ agreement expands coverage to 80.8% at 66.0% accuracy. When the strongest agreement is only 2/4, accuracy falls to 34.8%, well below the unanimous subset and only modestly above chance, indicating that model disagreement is a useful uncertainty signal, not random noise (Fig. 6a).

Human voting exhibits a steeper tradeoff. Expert unanimous agreement (among articles with at least two expert ratings) achieves 69.2% accuracy at 10.8% coverage, while $\geq 50\%$ junior consensus reaches 56.0% accuracy at 20.8% coverage. Without consensus filtering, junior and expert full-panel plurality accuracies are 40.0% and 39.2% on all 120 articles (Fig. 6b).

The contrast with frontier models is sharp. Cross-model voting using a diverse four-model frontier set (Gemini 3.1 Pro, Claude Opus 4.6, GPT-5.2 High, GLM-5) reached only 32.7% on the 104 articles with a clear plurality, and higher consensus thresholds did not recover performance (Fig. 6a,c). Frontier models fail to create meaningful consensus because post-training biases cause correlated errors across the same articles.

SFT cross-model consensus delivers a 15-percentage-point boost over the average single SFT model (56.9% to 72.0% at 4/4 consensus) because SFT models, despite independent training on different architectures, converge on a shared evaluative signal (inter-model Cohen's kappa = 0.50–0.60, versus human kappa approximately 0.03–0.05; Supplementary Table ST12). The practical implication is a selective evaluation workflow: route the 42% of submissions where all four SFT models agree (at 72% accuracy) directly, and concentrate reviewer attention on genuinely ambiguous cases.

Cross-field replication in economics

The results above establish the institutional trace mechanism, its mechanistic underpinnings, and its aggregation properties in management. A natural question is whether the mechanism is specific to that field or generalizes to other social science disciplines with distinct publication cultures and evaluation norms. We constructed an additional benchmark: 200 held-out research pitches in economics, balanced across the same four tiers and drawn from 2025 publications (Supplementary Methods SM9). We then trained new SFT checkpoints on field-specific institutional traces (5,593 economics pairs) using the same procedure.

The mechanism replicated (Fig. 7). The best single model (Qwen3-30B) reached 69.5% accuracy, with all three architectures at least 64% (Fig. 7a,b). Architecture-matched base models remained at chance (25–26%), confirming that the gains are attributable to fine-tuning (Supplementary Table ST15). Calibrated confidence replicated as well: accuracy rose sharply when coverage was restricted to high-confidence predictions (Fig. 7c).

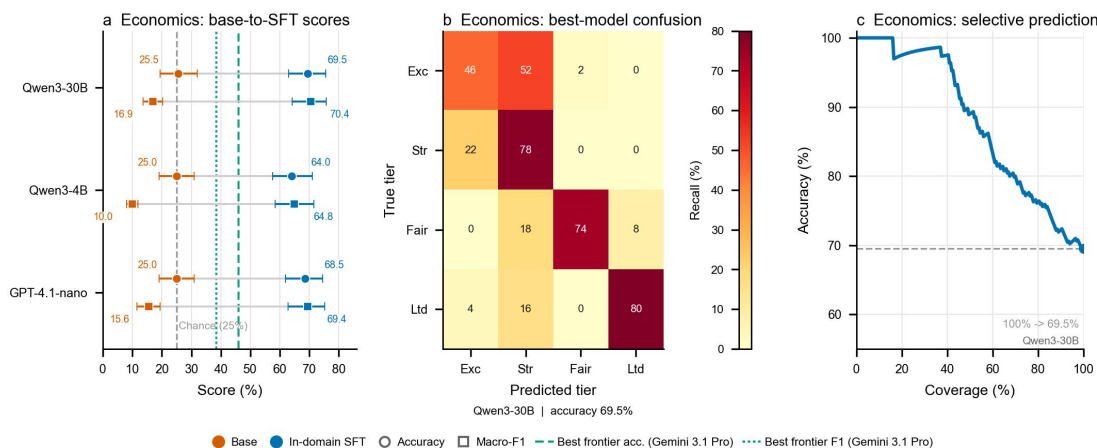


Figure 7 | Institutional-trace retraining replicates in economics

The economics validation reuses the same four-tier output space and the same `rq_with_context` representation logic as the main benchmark, but evaluates a held-out 2025 subject-specific sample ($N = 200$; 50 items per tier). **a**, Architecture-matched base-versus-in-domain-SFT scores in economics for Qwen3-30B, Qwen3-4B, and GPT-4.1-nano, shown as linked points with circles for accuracy and squares for macro-F1. Horizontal whiskers show bootstrap 95% confidence intervals recomputed from the raw subject prediction files. The grey dashed line marks the 25% chance baseline. The green dashed and dotted reference lines mark the best frontier (Gemini 3.1 Pro) full-set accuracy and macro-F1 in economics, respectively (46.0% accuracy; 38.45% macro-F1), under the same single-label frontier scoring convention used for the subject summaries. **b**, Row-normalized confusion matrix for the best single economics model (Qwen3-30B SFT), showing that the strongest in-domain checkpoint recovers all four tiers rather than collapsing into the middle categories. **c**, Selective-prediction curve for that same best economics model, with articles ordered from highest to lowest predicted confidence; the curve shows how accuracy rises as lower-confidence cases are deferred. Together, these panels show that the institutional-trace mechanism replicates in economics, while remaining clearly separated from the best currently available frontier comparator and retaining a practically useful confidence signal under the new subject-specific validation.

Strikingly, even the management-trained SFT GPT-4.1, which never encountered economics articles or journal tiers during training, achieved 43.5% on the economics benchmark ($p = 9.2 \times 10^{-9}$ versus chance), a +14.0 percentage-point gain over its own base model (29.5%, not significantly above chance; Supplementary Fig. 7). This cross-field transfer suggests that evaluative signals share structure across related social science disciplines: some component of the taste learned from management institutional traces generalizes to economics research evaluation. A model trained jointly on management and economics institutional traces maintained strong performance across both domains (Qwen3-30B: 69.5% economics, 61.7% management; Extended Data Fig. 8), confirming that evaluative signals from distinct fields do not interfere when combined.

Discussion

Human evaluators cannot reliably agree on which research ideas belong in which quality tier. Across nearly 3,000 ratings, experienced gatekeepers (editors and editorial board members at leading journals) showed categorical agreement barely above chance (Fleiss' $\kappa = 0.047$); among a larger panel of doctoral and postdoctoral researchers, the pattern was the same. This reflects not individual failure but the nature of the problem. Evaluative judgment in science is collective tacit knowledge[20]: no individual reviewer can articulate what distinguishes exceptional from merely competent work, yet the institutional system, integrating thousands of such judgments over decades, produces reliable quality stratification[19]. The knowledge is real, but it is not in any one person's head. It resides in the accumulated record of what was selected, funded, and published, an institutional trace that encodes a field's accumulated taste in a form no explicit rubric has managed to capture[8, 11].

The dominant push in AI for science assumes the bottleneck is generation. Autonomous systems can now produce complete scientific manuscripts at scale: Project APE has generated over 300 economics research papers using frontier models with multi-stage review pipelines[46], and the AI Scientist system has produced end-to-end machine-learning manuscripts submitted to peer-reviewed venues[47]. Yet in head-to-head tournaments against published human research, AI-generated papers win less than 5% of the time[46], a margin difficult to distinguish from noise in the judging process itself. AI tools have expanded scientific output but may have contracted its quality and focus[5]. These systems demonstrate that raw generative capability is not the constraint: frontier models can synthesize literature, run analyses, and write fluent prose. What they cannot do, and what our results confirm, is evaluate which ideas deserve pursuit in the first place. Frontier reasoning models capture barely 8% of the available headroom on our benchmark (Fig. 2a); supervised fine-tuning on institutional traces captures nearly half, at a fraction of the cost (Fig. 4a). The evaluative signal was never missing; it was never the training target.

These results suggest a practical alternative to autonomous scientific generation: an upstream evaluative filter, trained on institutional traces, that screens research ideas before resources are committed. Several features of the fine-tuned models make such deployment realistic. The models know when they are likely to be right: calibrated confidence concentrates the most reliable predictions into a high-precision subset that could be acted on directly, while flagging uncertain cases for human review (Fig. 5a,b). The evaluative capacity also generalizes beyond the training format. Without exposure to pairwise comparisons during training, the models transfer to head-to-head discrimination tasks, indicating learned quality ordering and not just memorized category assignments (Fig. 5c,d). Cross-model consensus provides a second reliability lever: when independently trained models agree, accuracy rises sharply; when they disagree, that disagreement itself becomes a useful triage signal (Fig. 6a). And because the SFT ensemble and expert panels err on largely different articles, combining the two could recover substantially more of the available headroom than either alone (Supplementary Fig. 5). The practical workflow this enables is selective: route clear cases through the model, concentrate scarce reviewer attention on the genuinely ambiguous remainder.

The economics replication strengthens these implications. The same fine-tuning procedure, applied to a different field with distinct journals, evaluation norms, and intellectual traditions, produced even stronger results (69.5% best single model; Fig. 7). Pooled training on both fields simultaneously preserved performance in each domain (Extended Data Fig. 8), suggesting that evaluative signals from different disciplines coexist without interference. A management-trained model that never encountered economics articles still exceeded chance on the economics benchmark by 14 percentage points (Supplementary Fig. 7), pointing to shared evaluative structure across related social sciences. These cross-field results shift the interpretation from a field-specific finding to a general mechanism: wherever institutional gatekeeping has operated long enough to leave a trace, that trace is likely learnable.

Several limitations qualify these conclusions. Our benchmarks cover two social science fields; whether the mechanism extends to STEM disciplines with different epistemic structures, where empirical reproducibility provides a partial quality signal absent in social science, remains untested. The transfer results under input compression reveal an open boundary condition: the larger fine-tuned model carries evaluative signal across radical input compression, but the smaller model does not (Extended Data Fig. 7), indicating that representational capacity governs transfer robustness. Future work should test which model properties or training-data characteristics determine this threshold. Institutional traces are a proxy for quality, not an objective ground truth. They encode the accumulated consensus of a gatekeeping system with well-documented biases: toward incremental over novel work[18], toward established methodologies, toward dominant paradigms, and toward a competitive bar that rises over time as submission volumes grow. The SFT model learns what the system historically rewarded, which is not identical to what is objectively best. This limitation is real but bounded: in domains where quality is irreducibly unverifiable,

institutional consensus over time is not a proxy for taste; it is the operational definition of taste[19, 46]. The model’s calibrated uncertainty provides an internal check on its own reliability that the system it learned from lacks. The signal is also not static: models trained on older institutional traces still outperform frontier systems but show drift in tier calibration as competitive thresholds shift (Extended Data Fig. 6), implying that periodic retraining will be necessary as institutional standards evolve.

The mechanism generalizes beyond any single discipline. In any domain where collective human evaluation has operated over time, the historical record of what was selected, funded, or rewarded constitutes a learnable signal: venture investing, where expert prediction is notoriously poor[32]; grant allocation, where reviewer agreement approaches zero[12]; and creative industries, where market outcomes consistently defy expert forecasts. The social and behavioral sciences, where evaluative judgment resists formal verification and submission volumes far outstrip reviewer capacity, stand to benefit most immediately. The approach is also uniquely cost-effective: SFT requires only historical decision records already deposited in institutional archives, total training cost was under \$300, and the resulting models provide calibrated confidence scores that make their uncertainty transparent and auditable. For science at scale, the path forward may not be AI systems that replace human judgment or that generate research autonomously, but systems that learn the taste human institutions have accumulated over decades and apply it where human bandwidth cannot reach. Autor predicted that machine learning would overcome Polanyi’s paradox, the barrier between knowing and telling, by learning from outcome data instead of explicit instruction[39]. Our results confirm this prediction: frontier models that receive evaluative criteria through prompts perform at chance; SFT that learns from outcome data recovers nearly half the available headroom in management and exceeds it in economics. Publication records function as implicit feedback from the evaluative community: individually noisy but collectively informative, encoding what Hinton et al. termed ‘dark knowledge’[38], evaluative structure embedded in institutional sorting but invisible in any stated criterion. Taste was never uniquely human; it was always deposited in the institutional record, waiting for a learning procedure simple enough to extract it.

Acknowledgements

We thank the 48 expert gatekeepers – editors and editorial board members at leading journals in organizational behavior and management – and the 174 doctoral and postdoctoral researchers who volunteered their time to evaluate research pitches. Their careful judgments provided the human benchmark against which all AI systems were measured, and their participation made this study possible.

Qiuping Peng and Liyun Zhang assisted with data collection and project management.

Data availability

The de-identified benchmark data, model prediction files, human-rating files, figures, tables, and reproducibility scripts supporting this study will be released through the project repository at <https://github.com/FutureTech-OB/ai-taste>. The repository will also provide the fine-tuning code, training data, and information on access to the resulting model weights.

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Methods

Study design. Evaluative judgment in science is widely recognized as tacit and institutionally distributed[8, 16, 20]: ethnographic evidence shows that even experienced gatekeepers rely on intuitive assessment and disciplinary sensibility rather than rule application[16], and decades of structured rubric design have not improved inter-rater reliability[11]. This study was designed to test whether AI systems can recover such judgment from institutional decision traces when asked to assess research ideas before empirical results are known. We applied the protocol to two social science fields, organizational psychology/management and economics, to test both the mechanism and its generality. We centered the protocol on early-stage research-pitch evaluation instead of full-paper critique, and imposed a shared four-tier decision space across all evaluator classes (human and AI) so that performance differences reflect differences in judgment, not task format, criterion framing, or access to stylistic cues. Human evaluation, frontier-model benchmarking, and mechanism tests (reinforcement-learning ablation, pairwise discrimination, voting consistency) were conducted in management; economics served as a replication of the core SFT mechanism with architecture-matched base controls.

Benchmark construction and rationale for labels. We constructed balanced benchmarks in two fields: 120 article-derived research pitches in organizational psychology and management (30 per tier, 19-journal source universe; Supplementary Methods SM5) and 200 in economics (50 per tier, 38-journal source universe; Supplementary Table ST14). All source articles were published after mid-2025. In both fields, the four tiers (exceptional, strong, fair, and limited) were derived from journal-level publication outcomes mapped to pre-specified tier frameworks reviewed by domain experts. Operationally, exceptional denotes elite field-defining journals, strong near-elite specialty outlets, fair established field journals, and limited lower-prestige or narrower-scope outlets. Journal-level labels were chosen because they represent the most stable institutional record of long-run gatekeeping decisions currently available at scale[19, 20]: individual reviewer judgments are dominated by noise[30], but the system-level sorting that produces prestige-tier differentiation integrates iterated editorial consensus over time. This makes the training target explicit: the model learns to recover collective institutional judgment, not to approximate any single reviewer’s opinion.

A balanced design was chosen to prevent models from exploiting class-frequency priors and to enable interpretable per-tier comparisons, especially at the quality extremes where editorial decisions are most consequential. Tier balance is ensured by construction (30 pitches per tier in management; 50 per tier in economics).

Input standardization and extraction workflow. Evaluator comparisons are meaningful only if all evaluators see the same information. Each source article was therefore transformed into a standardized research-pitch text using a fixed extraction workflow (Supplementary Methods SM4). These pitches presented the core research question and theoretical framing without detailed methods, full empirical findings, journal identity, or author identity. The primary extractor was Qwen3-235B-A22B-Instruct, a large language model selected for extraction quality and format stability after cross-validation against alternative strong models. We generated five textual formulations (varying in the balance of theoretical context, research-question specificity, and methodological detail) and selected the research-question-with-context form as the primary representation through pilot evaluation on a held-out development set, because it preserves theoretical motivation while stripping later-stage evidence that would trivially encode publication outcomes. This normalization isolates idea-level assessment and reduces shortcut learning. The same extraction pipeline and policy were applied across both fields, all model families, and materials given to human raters so that evaluator comparisons remained behaviorally aligned.

Training corpus and leakage control. For each field, supervised training data were drawn from the field’s source universe and fully disjoint from its held-out benchmark. The management corpus comprised a primary recent/new slice (4,479 research-pitch/journal-outcome pairs from 19 journals) and an older slice for temporal comparison (3,368 pairs). The economics corpus comprised 5,593 pairs from 38 journals. A pooled corpus combining both fields (~10,072 pairs) was also constructed. Tier labels followed the same journal-level mapping used in evaluation. Ambiguous samples were excluded during curation to reduce avoidable label noise in the training target.

Temporal separation was introduced to control contamination risk: source articles underlying the benchmark pitches were published after mid-2025, whereas the base models used for fine-tuning were released before that period.

Because publication outcomes are influenced by factors beyond idea quality (execution quality, writing quality, reviewer-assignment effects, editorial fit) and because the model input captures only idea-level information, a noise ceiling is expected by design[30] (Supplementary Methods SM8). This ceiling affects absolute accuracy interpretation while leaving relative comparisons internally valid, since the same information constraint is imposed across all evaluator classes.

Evaluation criteria and prompt freezing strategy. All evaluators used the same two-axis criteria framework centered on originality and usefulness, consistent with editorial guidance literature in management research[9, 10]. A shared rubric was imposed on both humans and AI so that score differences would reflect evaluator behavior, not criterion mismatch. For economics, the same framework was adapted to a broader social-science framing while preserving the tier structure and zero-shot design (Supplementary Methods SM9). Tier definitions and prompt variants are provided in Supplementary Methods SM6.

Three prompt variants were pre-specified before final runs: an expert-anchored rubric (using the language and standards of experienced journal editors), a simplified plain-language rubric, and a journal-anchored rubric (referencing specific publication venues as quality anchors). The expert-anchored and journal-anchored variants yielded comparable frontier performance in pre-analysis checks, but journal-anchored prompts induced models to enumerate journal lists and hunt for venue-specific stylistic cues during reasoning; the expert-anchored variant was therefore frozen as the primary prompt to avoid this confound, making downstream comparisons conservative for frontier systems instead of optimized for fine-tuned models. Cross-model prompt sensitivity analysis is reported in Extended Data Fig. 1. All evaluations used zero-shot prompting to keep conditions comparable across evaluator classes, including models and human raters, and to avoid noisy exemplar anchoring from publication outcome labels. Unresolved model outputs were coded as incorrect for fixed denominator reporting (overall non-compliance rate <1% across 10,560 individual runs).

Human evaluation protocol. The human study was approved by institutional review board review (Project No. THU-04-2026-0034).

Two panels were used to separate experienced gatekeeper judgment from trainee judgment under the same task definition. The expert panel comprised 48 raters contributing 384 ratings (8 research pitches per rater; mean 3.2 ratings per pitch), recruited through personal professional networks to ensure high domain relevance and intrinsic motivation; two anonymous response identifiers were reconciled as two distinct raters. The junior panel comprised 174 doctoral and postdoctoral researchers contributing 2,530 ratings (mean 14.5 pitches per rater; mean 21.1 ratings per pitch), recruited via doctoral and postdoctoral networks; junior raters who spent fewer than one minute per pitch were excluded to remove perfunctory responses. Unfiltered expert data were prespecified as the primary expert analysis because filtering reduces pitch-level vote counts and increases tie sensitivity, limiting the pitches on which majority voting can be computed. Filtered sensitivity analyses are reported in Supplementary Table ST7.

For each benchmark pitch, raters reported prior exposure (yes/no), tier assignment using field-familiar labels (Top, Top-, Good, Fair, mapped to exceptional, strong, fair, limited), confidence (5-point Likert scale), and topic familiarity (5-point Likert scale). These labels mirror common journal-evaluation shorthand, while all analyses convert them deterministically to the unified four-tier labels used in the AI protocols and manuscript. The full survey instrument and procedural details are provided in Supplementary Methods SM7. We collected confidence and familiarity because prior work shows that peer-review agreement and evaluator quality are weakly coupled to conventional expertise signals[11, 12]; these variables allowed direct testing of whether the same pattern held in our dataset. Demographic and background summaries are in Supplementary Table ST5; individual expert accuracy values are in Supplementary Table ST9; matched-N Monte Carlo details are in Supplementary Table ST10; and a concise prior-exposure descriptive summary is reported in Supplementary Table ST11.

AI model families and inference protocols. We evaluated three AI families under one frozen instruction scaffold. The frontier cohort included 11 reasoning models: Gemini 3.1 Pro, Claude Opus 4.6, GPT-5.2 High, Gemini 2.5 Pro, Qwen 3.5 Plus, DeepSeek V3.2, Seed 2.0 Pro, MiniMax M2.5, Kimi K2.5, Grok 4.1 Fast, and GLM-5. Gemini 3.1 Pro is retained with a contamination-risk caveat because some reasoning traces partially reproduced benchmark content. Each frontier model was sampled eight times per pitch to account for the inherent randomness in text generation. The primary frontier metric was pitch-mean eight-sample accuracy. For discrete diagnostics, we used per-pitch majority vote from the same eight runs and reported effective sample size after tie exclusion. Protocol coverage is summarized in Supplementary Table ST13.

We additionally evaluated chat/log-probability tracks (GPT-5.2 chat, Kimi K2 chat, DeepSeek Chat) and architecture-matched base controls (the same underlying models before any fine-tuning). Each base model was evaluated on the full 120-pitch benchmark using the same frozen prompt and four-label log-probability classification as its SFT counterpart, providing architecture-matched controls that isolate the effect of fine-tuning from pre-existing model capability. Non-thinking models (base, chat, and supervised fine-tuned checkpoints) were evaluated by four-label token log-probability classification, which yields deterministic class probabilities and avoids free-text parsing failure modes. The same log-probability protocol was used for all four SFT checkpoints and their six pairwise probability-averaging ensembles (Supplementary Table ST4).

This dual protocol reflects a difference in what each API surface exposes: reasoning APIs produce stochastic natural-language completions best stabilized by repeated sampling, while non-reasoning APIs expose token-level log-probabilities that yield calibrated label distributions in a single deterministic call, enabling both confidence analysis and direct probabilistic comparison. Cost and protocol comparisons are summarized in Supplementary Table ST2.

Supervised fine-tuning. Supervised fine-tuning (SFT) is a procedure in which a pre-trained language model is further trained on a curated dataset of input–output examples to specialize it for a particular task[41]. We fine-tuned base models spanning two model families and multiple scales using the same procedure in both fields. In management, four models were trained: GPT-4.1, GPT-4.1-nano (a smaller, more efficient variant), Qwen3-4B-Instruct, and Qwen3-30B-A3B-Instruct (a mixture-of-experts architecture with 30 billion total parameters but only 3 billion active at any time). In economics, three of these four architectures were trained (GPT-4.1-nano, Qwen3-4B, Qwen3-30B); GPT-4.1 was omitted because its API fine-tuning cost was disproportionate to the three cost-effective architectures sufficient to demonstrate cross-architecture replicability. Pooled models were additionally trained on the combined corpus from both fields (~10,072 pairs). The cross-family, cross-scale, cross-field design tests whether the evaluative signal is recoverable from institutional traces generally or specific to a particular architecture, scale, or discipline[34].

Each training example consisted of the frozen evaluation prompt wrapping a research-question-with-context pitch as input, with a single tier-label token (one of four: exceptional, strong, fair, limited) as the completion target. Labels were designed as semantically descriptive single tokens to ensure distinguishability under first-token log-probability classification and to leverage the models’ pre-trained quality grading representations. The same prompt template was used for training and evaluation, eliminating prompt mismatch confounds. Given input text x and label $y \in \{1, 2, 3, 4\}$, training minimized label-token negative log-likelihood, a standard objective that increases the probability the model assigns to the correct tier label:

$$\mathcal{L}_{\text{NLL}}(\theta) = -\frac{1}{N} \sum_{i=1}^N \log p_{\theta}(y^{(i)} | x^{(i)}),$$

with loss computed on label tokens only and input tokens masked from gradient updates. Input masking is critical: it prevents the model from memorizing prompt tokens and forces the model to learn exclusively the mapping from research content to quality tier. Training corpus sizes are reported above (Training corpus and leakage control). Management training cost was below \$300 USD across all four models, with the largest single cost arising from GPT-4.1 API fine-tuning (~\$200); GPT-4.1-nano required ~\$10 via API, while the open-source Qwen3-4B and Qwen3-30B-A3B checkpoints required approximately one and eight A100 GPU-hours respectively; economics training costs were comparable for the three architectures used (Supplementary Table ST2). Qwen checkpoints were trained locally using TRL; GPT checkpoints were trained via the OpenAI fine-tuning API. All models used near-default settings without hyperparameter search; full settings are reported in Supplementary Methods SM1. Consistent performance gains across both a proprietary black-box pipeline and a fully controlled open-source pipeline, across two fields with independent journal hierarchies, isolate the training signal and not any particular optimization decision as the causal factor.

To test out-of-format generalization under severe information compression, we also evaluated a compressed-input transfer set built from the same held-out 120 articles but using only the single-sentence idea-statement field (core_rq_short) extracted in Supplementary Methods SM4. Unlike the primary full idea-summary representation (rq_with_context), this one-sentence version removes almost all theoretical motivation and contextual framing, leaving only the focal research question in one sentence. No checkpoint was trained on this format: all supervised fine-tuning used the fuller idea-summary input only, so this analysis tests transfer at evaluation only, not in-format test performance. The four GPT-family base/SFT checkpoints were scored with the same four-label log-probability classifier used elsewhere. Predictions were recovered by argmax over the four label log-probabilities; any missing label was treated as negative infinity; and exact ties were resolved by a fixed label order (exceptional, strong, fair, limited). Because the benchmark remains exactly balanced across four tiers, descriptive chance performance is 25%

accuracy, and one-sided exact binomial tests versus 25% were used only as a compact above-chance diagnostic for this auxiliary transfer evaluation (Extended Data Fig. 7). This auxiliary analysis was interpreted as a test of generalization robustness rather than as a replacement for the primary in-format benchmark comparison.

A two-model ensemble was built by averaging softmax-normalized label probabilities from GPT-4.1-nano (SFT) and Qwen3-30B-A3B (SFT), then selecting the maximum probability class with deterministic resolution of any exact class probability ties. We evaluated all six pairwise SFT ensembles on the held-out benchmark and ranked them by accuracy, then macro F1, with a fixed model order convention used only if those metrics remained tied; under this rule GPT-4.1-nano + Qwen3-30B-A3B was retained as the primary pair (Supplementary Table ST4). Because all six pairwise ensembles exceeded the frontier average by 28.1 to 29.8 percentage points, the core finding is robust to ensemble composition and not dependent on a single fortuitous combination.

Architecture-matched base controls serve as the primary comparison for isolating the SFT effect in both fields: because each fine-tuned model is evaluated against its own pre-training checkpoint on the identical benchmark, observed gains are attributable to the fine-tuning procedure and not to differences in base-model capability or architecture. GPT-4.1 was additionally retained for the cross-field transfer evaluation (management-trained model tested on the economics benchmark; Supplementary Fig. 7) because it demonstrated the strongest generalization capacity under input compression (Extended Data Fig. 7). Full economics journal-to-tier mapping details are in Supplementary Methods SM9 (Supplementary Table ST14).

Reinforcement-learning ablation. To test whether explicit reasoning optimization improves this task beyond supervised alignment, we trained reasoning-enabled Qwen3-4B and Qwen3-32B checkpoints with a modified GRPO-style objective[28, 31, 40] (Supplementary Methods SM2–SM3). The implementation removed the KL penalty, used token-level normalization, and adopted asymmetric clipping to favor positive-advantage updates. We also introduced adaptive privileged sampling for low-accuracy items[42, 43] to maintain reward contrast during training.

RL checkpoints were evaluated using the same 8-run sampling protocol as frontier models, with run-pooled and pitch-mean eight-sample accuracy reported alongside non-tied majority vote. RL training inherently requires reasoning-enabled model configurations that generate explicit chain-of-thought before a final label; this is not a confound but the mechanism under test. The comparison asks whether explicit deliberation aids or impedes evaluative accuracy, a question motivated by converging evidence from cognitive science that verbal articulation can degrade holistic judgment[35], and from machine learning that chain-of-thought prompting yields negligible or negative gains on tasks outside mathematics and symbolic reasoning[36, 37]. Reinforcement learning from human preferences has additionally been shown to instill agreement-seeking behavior[28, 29] and reduce output diversity through mode collapse, and GRPO-style methods represent the current state of the art for reasoning optimization[31]. This RL track was therefore treated as a mechanism test, not a deployment candidate: if explicit chain-of-thought policy optimization cannot match direct supervised alignment on this task, the implication is that the evaluative signal resides in the institutional traces themselves and not in reasoning architecture.

Pairwise discrimination experiment. We designed a label-free pairwise task to distinguish gains in intrinsic discrimination from gains in rubric alignment. Each trial presented two research pitches sampled from different tiers, and the model selected the stronger one without seeing tier labels or rubric text, testing whether a model can tell which of two ideas is better even without the four-category framework. We used a fixed 300-pair stratified set sampled from the 120 benchmark pitches with fixed random seed 32 (150 pairs at tier distance 1, 100 at distance 2, and 50 at distance 3, where distance 1 means adjacent tiers and distance 3 means the most separated tiers).

The narrated pairwise comparison set was the shared four-model subset used across the main text, Fig. 5, Extended Data Fig. 2, and Supplementary Table ST1: SFT GPT-4.1, Gemini 3.1 Pro, GPT-5.2 High, and the GPT-4.1 baseline. Because this task removes absolute category assignment, improvements here are interpretable as improvements in relative quality discrimination rather than prompt-following alone. Fig. 5 carries the headline pairwise comparison for this same subset: overall weighted accuracy with exact McNemar tests and the hard-boundary summary. Extended Data Fig. 2 retains the six-pair-type heatmap and discordance decomposition for the same subset.

Voting-consistency analysis. Consensus analyses were prespecified to quantify how reliability changes with stricter agreement thresholds across evaluator classes. For SFT, we analyzed 4/4, 3/4, and 2/4 cross-model agreement policies. For humans, we analyzed junior vote share thresholds and expert unanimity subsets. For frontier models, we analyzed both within-model repeated-sampling aggregation and fixed cross-model voting on a diverse subset of four top models (Gemini 3.1 Pro, Claude Opus 4.6, GPT-5.2 High, GLM-5).

Inter-rater consistency used Fleiss' κ for human panels and pairwise Cohen's κ for model-model agreement. Majority-vote results excluded tied pitches and always reported effective non-tied N. Ties were not broken randomly because random resolution introduces avoidable variance and obscures whether an evaluator family is genuinely indecisive on difficult cases. Full agreement diagnostics are reported in Supplementary Table ST12.

Statistical analysis. All analyses were run in Python (NumPy, SciPy, pandas). The primary endpoint was four-class exact-match accuracy. Secondary endpoints included macro-F1, per-tier precision/recall/F1, confusion matrices, and calibration diagnostics.

Ordinal inter-rater agreement was assessed with Krippendorff's alpha. Paired evaluator comparisons used McNemar tests on paired correctness vectors. For the pairwise task summarized in Fig. 5 and diagnosed further in Extended Data Fig. 2, significance was computed with the two-sided exact McNemar/binomial test on item-level discordant pairs and reported as raw unadjusted P values for the plotted SFT comparisons versus Gemini 3.1 Pro, GPT-5.2 High, and GPT-4.1 baseline. Frontier-cohort heterogeneity was tested with Cochran's Q. Ordinal or non-Gaussian analyses used Spearman correlation, Mann-Whitney U, and Kruskal-Wallis tests. Individual-level comparisons tested whether the SFT ensemble accuracy differed from the distribution of individual human evaluator accuracies using a one-sample t-test with the ensemble score as the reference value. Confidence intervals were estimated by bootstrap resampling (10,000 draws). Multiple-testing corrections were used only where explicitly stated for broader evaluator families and prompt/model sweeps; the focused pairwise analyses in Supplementary Table ST8 and Fig. 5 / Extended Data Fig. 2 report raw unadjusted P values.

The reported significance tables highlight four comparator rows: frontier average (11 models), best frontier model under the conservative protocol (Gemini 3.1 Pro), expert majority vote, and junior majority vote. The paired primary comparisons emphasized in the manuscript and supporting tables are SFT ensemble versus expert majority vote and versus the best frontier model under the conservative protocol; frontier-average and junior-majority rows are reported as additional headline secondary comparisons. Additional paired evaluator tests were reported as secondary analyses, with raw unadjusted P values shown in the supporting tables and any multiplicity-adjusted interpretation stated explicitly in text. Headroom is defined as $(\text{accuracy} - \text{chance}) / (100\% - \text{chance})$, where chance = 25%, giving the fraction of improvable performance captured by a given evaluator. Human judgment analyses incorporated the noise framework of Kahneman et al.[30] to interpret the dissociation between low categorical agreement and moderate ordinal agreement. Majority-vote comparisons involve reduced effective N because tied pitches are excluded, which limits statistical power relative to individual-level tests; this is particularly relevant for the comparison between SFT and expert majority vote, where tie exclusion reduces the evaluable set.

Calibration for probabilistic evaluators used expected calibration error (ECE) and Brier decomposition; selective prediction was evaluated by plotting accuracy as a function of coverage under confidence threshold sweeps. Monte Carlo matched-N analyses (5,000 random draws) were used to compare junior and expert majority voting at equivalent panel sizes, drawing expert-sized panels from the junior researcher pool and computing majority-vote accuracy on each draw. Label normalization across source-article metadata, human surveys, and model outputs used deterministic mapping rules (Supplementary Table ST6).

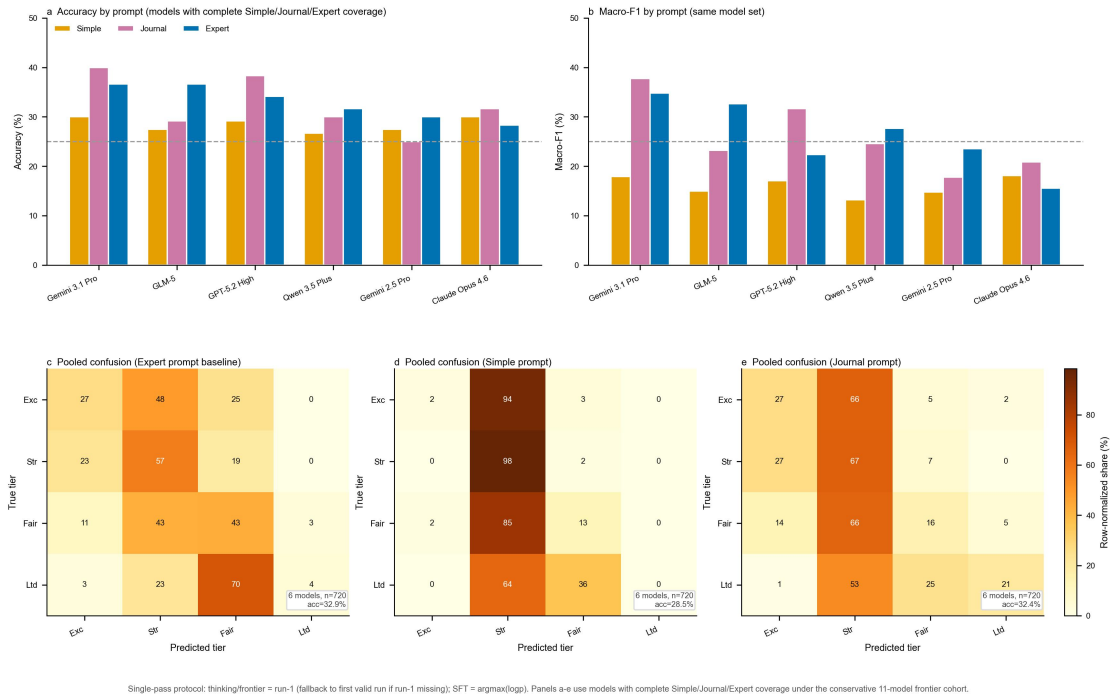
Data and code availability. All preprocessing rules, prompts, model inventories, hyperparameters, and sensitivity analyses are documented in Supplementary Methods and Tables. The benchmark split was fixed before final model comparisons, and label normalization rules were deterministic across source-article metadata, human surveys, and model outputs. Analysis code, processed benchmark files, prompts, and table-generation scripts will be released alongside the manuscript. Human data will be de-identified before release in accordance with approved ethics procedures.

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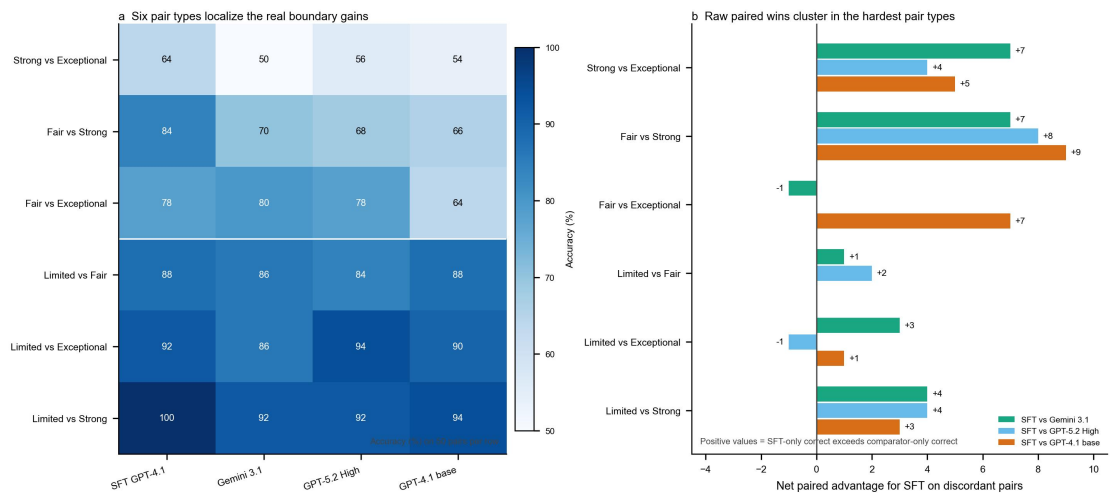
Extended Data

Extended Data Figures



Extended Data Figure 1 | Cross-model prompt-sensitivity landscape

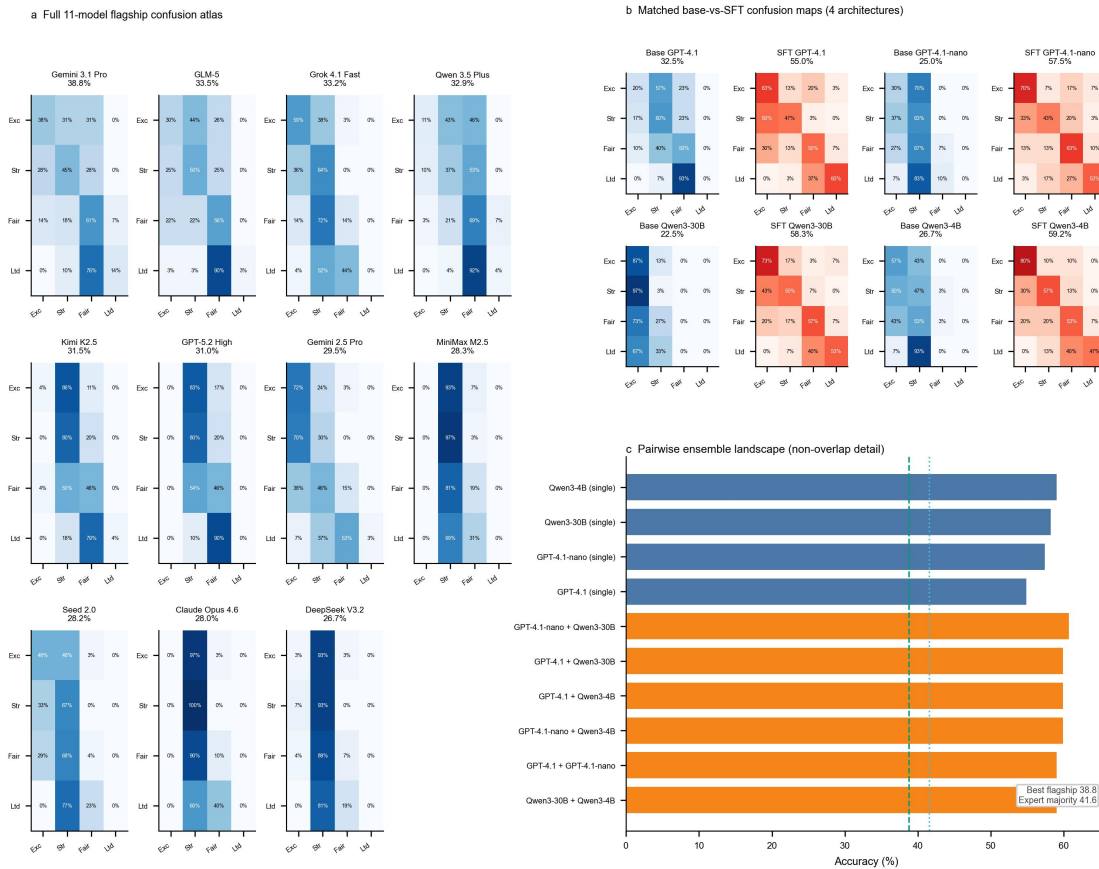
a, Accuracy across Simple, Journal-anchored, and Expert prompts for the six frontier models evaluated under all three prompt formulations in the conservative 11-model cohort: Gemini 3.1 Pro, GLM-5, GPT-5.2 High, Qwen 3.5 Plus, Gemini 2.5 Pro, and Claude Opus 4.6. Accuracy error bars show 95% binomial confidence intervals. **b**, Macro-F1 across the same prompt conditions, shown as a paired robustness metric to distinguish raw accuracy from balanced tier discrimination. **c**, Pooled row-normalized confusion matrix under the Expert prompt baseline. **d**, Pooled row-normalized confusion matrix under the Simple prompt. **e**, Pooled row-normalized confusion matrix under the Journal-anchored prompt. Together these panels show that prompt wording changes the shape of frontier-model collapse but does not recover robust four-tier discrimination. Because the Simple and Journal conditions use single-pass prompt evaluations whereas the Expert condition uses the conservative majority-based frontier protocol, cross-prompt differences should be interpreted directionally rather than as perfectly protocol-matched estimates.



Extended Data Figure 2 | Pairwise intrinsic discrimination details

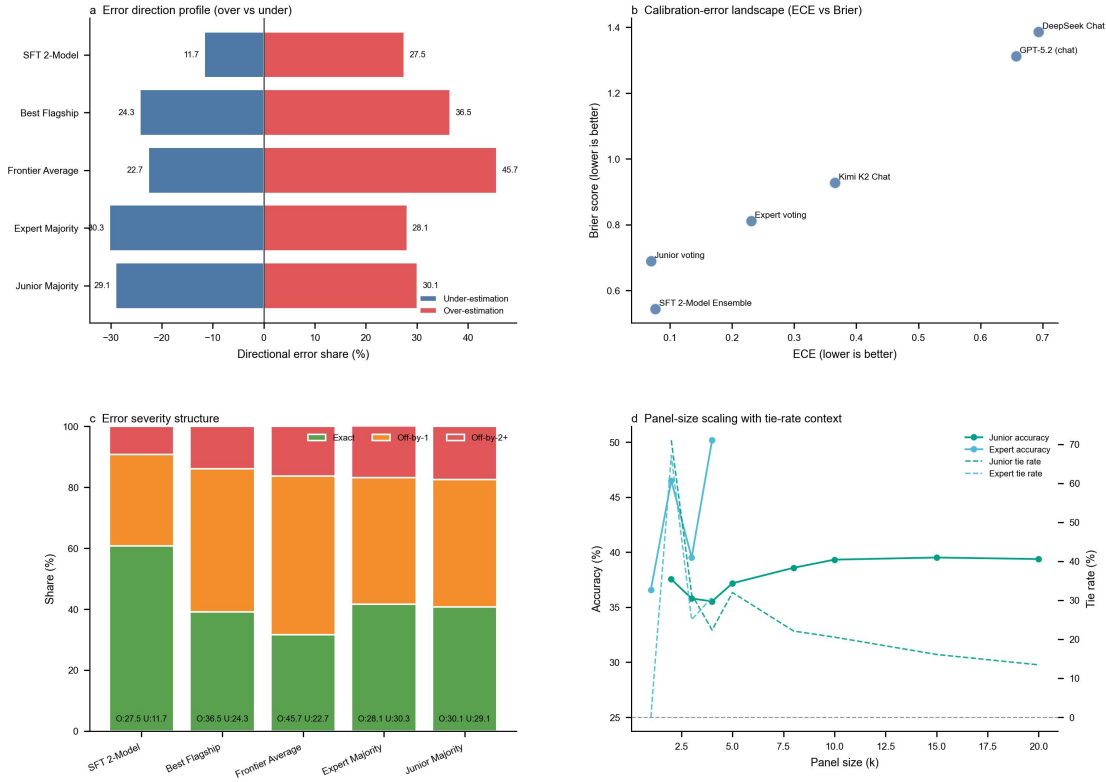
Models choose the stronger research pitch in pairwise head-to-head comparisons without tier labels or rubric text, isolating relative quality discrimination from absolute category assignment. The plotted ED2 comparator set is the same shared four-model subset used in Fig. 5 and Supplementary Table ST1: SFT GPT-4.1, Gemini 3.1 Pro, GPT-5.2 High, and the GPT-4.1 baseline. Fig. 5 now carries the headline overall-accuracy and hard-boundary pairwise panels. **a**, A six-pair-type heatmap (strong_exceptional, fair_strong, fair_exceptional, limited_fair, limited_exceptional, limited_strong) shows where performance gaps actually sit once

the broad distance bins are unpacked. **b**, Net discordant-pair decomposition (SFT-only correct minus comparator-only correct) shows that the paired significance is concentrated in the hardest pair types rather than in the near-ceiling easy pairs. Raw unadjusted exact McNemar P values for SFT GPT-4.1 versus Gemini 3.1 Pro, GPT-5.2 High, and the GPT-4.1 baseline are reported in data/statistics/S14_ED2PairwiseRawPValues.json.



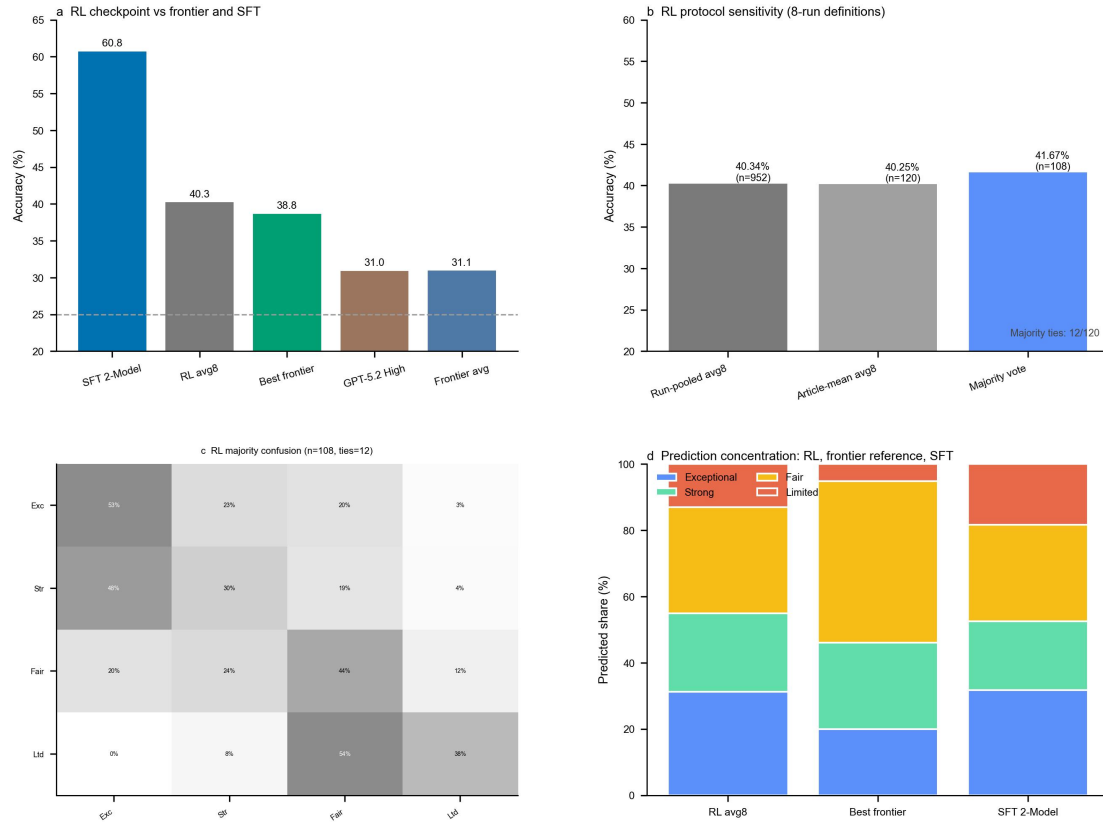
Extended Data Figure 3 | Full confusion diagnostics and SFT ensemble detail

a, The complete confusion-matrix atlas for all 11 frontier flagships extends the selected examples shown in Fig. 2 and makes the full range of collapse modes visible in one view. **b**, Architecture-matched base-versus-SFT confusion matrices show how supervised fine-tuning changes tier discrimination within each model family, shifting mass toward the diagonal rather than toward one or two dominant tiers. **c**, Accuracy ranking across the four single SFT models and six two-model ensembles shows that the ensemble advantage is robust across pair choices rather than tied to a single combination; dashed reference lines mark the best flagship frontier model and expert-majority performance for context.



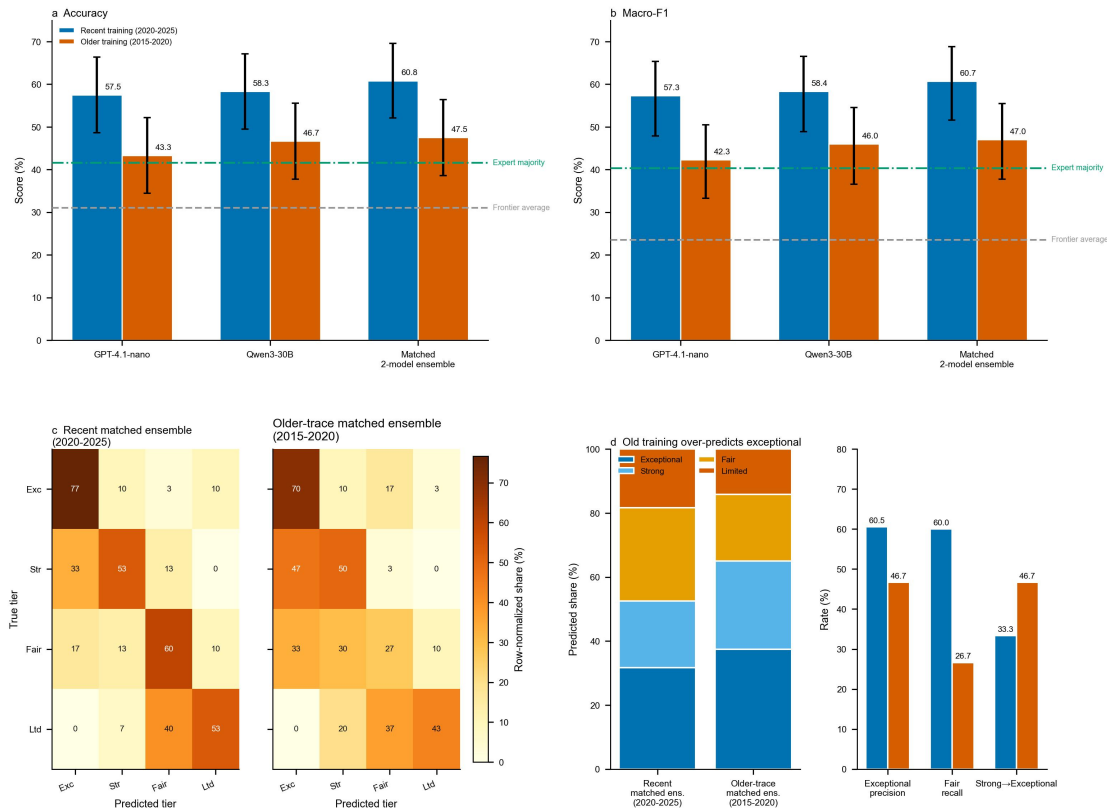
Extended Data Figure 4 | Calibration, error structure, and panel-size reliability

a, Directional error profiles compare under-estimation and over-estimation across the SFT ensemble, best frontier model, frontier average, expert majority vote, and junior majority vote, revealing that frontier systems are especially prone to over-estimating article quality. **b**, The calibration landscape compares expected calibration error (ECE) with Brier score across evaluators, with better-calibrated systems appearing closer to the lower-left corner. **c**, Error severity is decomposed into exact matches, off-by-1 errors, and off-by-2+ errors, with compact under- versus over-estimation summaries printed for each evaluator. **d**, Junior panel-size scaling is shown together with junior tie-rate trajectories and expert anchor lines for both accuracy and tie rate, illustrating that aggregation reduces indecision but plateaus in performance rather than eliminating the structural ceiling.



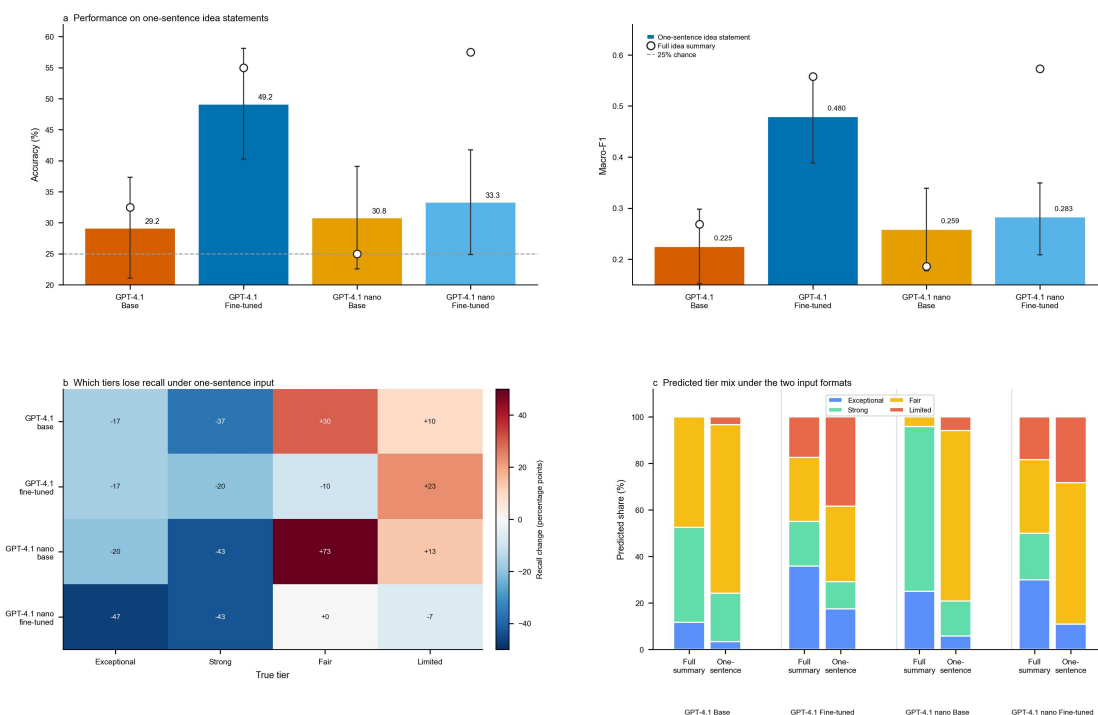
Extended Data Figure 5 | RL checkpoint diagnostics

a, RL checkpoint performance is compared with the SFT 2-model ensemble, the best frontier model, GPT-5.2 High, and the frontier average, locating RL between clean frontier baselines and supervised fine-tuning. **b**, RL performance is disaggregated into run-pooled eight-sample accuracy, pitch-mean eight-sample accuracy, and non-tied majority-vote accuracy, showing that the ranking is stable across reporting conventions. **c**, The RL majority-vote confusion matrix localizes residual errors by tier and shows that middle-tier separation remains the main weakness. **d**, Predicted-tier distributions compare RL with the best frontier model and the SFT 2-model ensemble, clarifying that RL reduces but does not eliminate collapse relative to the strongest frontier baseline.



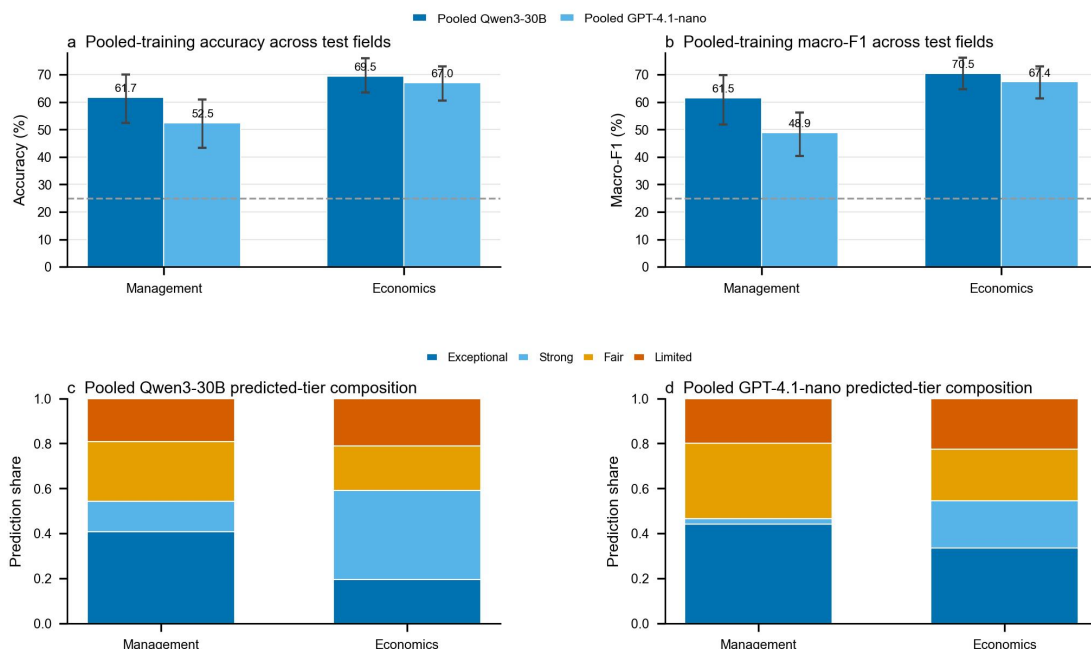
Extended Data Figure 6 | Temporal persistence of institutional traces

a, Accuracy for the matched recent-training slice (2020-2025) and older-training slice (2015-2020) on the two shared architectures (GPT-4.1-nano, Qwen3-30B) plus their matched 2-model ensemble. The benchmark itself is built from post-June-30-2025 articles, so the older slice introduces an intentional five-year training-time lag. Accuracy bars show binomial 95% confidence intervals, and dashed reference lines mark the frontier-average and expert-majority comparators from the benchmark summary tables. **b**, Macro-F1 for the same matched singles and matched 2-model ensemble, with bootstrap 95% confidence intervals and the same benchmark anchors, showing that temporal decay persists on a balanced discrimination metric rather than only on raw accuracy. **c**, Row-normalized confusion matrices compare the recent and older-trace matched ensembles, showing that the older trace retains exceptional-tier diagonal structure but loses substantially more fair-tier recovery and shifts mass upward. **d**, Predicted-tier distributions plus three targeted diagnostics quantify the older-trace drift: more exceptional predictions, lower exceptional-tier precision, lower fair-tier recall, and more strong → exceptional confusions.



Extended Data Figure 7 | Transfer from richer supervision to one-sentence core-question inputs

All fine-tuned checkpoints in this figure were trained only on the fuller research-idea summary and were never trained on the one-sentence idea-statement format. **a**, Overall performance on the one-sentence idea statement benchmark for architecture-matched GPT-family base and fine-tuned models, shown as accuracy and macro-F1 on the same 120 held-out articles. Bars show one-sentence-input results with binomial 95% accuracy confidence intervals and bootstrap 95% macro-F1 confidence intervals; hollow markers show the same model's performance on the fuller idea-summary benchmark, and the dashed line marks the 25% chance baseline. **b**, Per-tier recall change (one-sentence input minus full idea summary) localizes where compression hurts. GPT-4.1 fine-tuning retains substantial signal but loses recall in the exceptional, strong, and fair tiers while gaining limited-tier recall, whereas GPT-4.1 nano fine-tuning collapses on the strong tier entirely under compressed input. **c**, Predicted-tier distributions under fuller versus compressed inputs show that degradation is not a single generic middle-tier collapse: both base models concentrate heavily in the middle, GPT-4.1 fine-tuning becomes markedly more conservative and limited-heavy, and GPT-4.1 nano fine-tuning stops emitting strong-tier predictions on the compressed input.



Extended Data Figure 8 | Pooled multi-field training predicts management and economics

Pooled multi-field checkpoints trained jointly on management and economics institutional traces are tested separately on the two held-out field benchmarks. The test surface contains 120 management items and 200 economics items, with tier balance preserved in each field subset. **a**, Accuracy across the two test fields for the pooled Qwen3-30B and pooled GPT-4.1-nano checkpoints. The pooled Qwen3-30B achieves 61.7% on management – comparable to the 60.8% single-field SFT ensemble reported in the main text – and 69.5% on economics, matching the best in-domain economics SFT. The pooled GPT-4.1-nano reaches 52.5% on management and 67.0% on economics. Bars show point estimates with bootstrap 95% confidence intervals; the dashed line marks the 25% chance baseline. **b**, Macro-F1 across the same field-wise evaluations (Qwen3-30B: 0.615 management, 0.705 economics; GPT-4.1-nano: 0.489 management, 0.674 economics), with bootstrap 95% confidence intervals, confirming that the larger pooled model maintains substantially stronger balanced discrimination than the smaller model across both fields. **c**, Predicted-tier composition for the pooled Qwen3-30B checkpoint across management and economics, shown as 100% stacked bars over the four tiers. **d**, Predicted-tier composition for the pooled GPT-4.1-nano checkpoint. Together, these panels demonstrate that evaluative signals from management and economics do not interfere when combined in training: the pooled Qwen3-30B matches or exceeds single-field benchmarks in both domains, while the capacity gap between the larger and smaller architecture shapes how balanced cross-field discrimination remains. GPT-4.1 was selected for the cross-field transfer analysis (Supplementary Fig. 7) because it demonstrated the strongest generalization capacity under input compression (Extended Data Fig. 7), retaining evaluative signal when full idea summaries were reduced to one-sentence inputs; this capacity made it the natural candidate for testing whether management-trained evaluative representations transfer to a different disciplinary domain.

Extended Data Tables

Extended Data Table 1 | Base model controls

Model	Model Key	N	Accuracy (%)	Macro F1	95% CI Lower (%)	95% CI Upper (%)	Above Chance (pp)	Headroom Skill (%)	Acc exceptional (%)	Acc strong (%)	Acc fair (%)	Acc limited (%)	Pred exceptional (n)	Pred strong (n)	Pred fair (n)	Pred limited (n)
Base (GPT-4.1)	gpt-4.1	120	32.5	0.268	24.2	40.8	7.5	10.0	20.0	60.0	50.0	0.0	14	49	57	0
Base (Qwen3-4B)	qwen3-4b	120	26.7	0.195	19.2	34.2	1.7	2.2	56.7	46.7	3.3	0.0	47	71	2	0
Base (GPT-4.1-nano)	gpt-4.1-nano	120	25.0	0.186	17.5	33.3	0.0	0.0	30.0	63.3	6.7	0.0	30	85	5	0
Base (Qwen3-30B)	qwen3-30b-a3b	120	22.5	0.112	15.0	30.0	-2.5	-3.3	86.7	3.3	0.0	0.0	97	23	0	0

Supplementary Information

Machines acquire scientific taste from institutional traces

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- SM7: Human Study Design
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- ST1: Pairwise Discrimination by Tier Distance
- ST2: Cost and Inference Regime Comparison
- ST3: Core Per-Class Metrics
- ST4: All Pairwise SFT Ensemble Combinations
- ST5: Human Panel Composition and Descriptives
- ST6: Label Normalization
- ST7: Filtering Sensitivity
- ST8: Pairwise McNemar Test Compendium
- ST9: Individual Expert Accuracy Distribution
- ST10: Monte Carlo Matched-N Analysis
- ST11: Prior-Exposure Descriptive Summary
- ST12: Agreement and Consensus Diagnostics
- ST13: Model Inventory and Access Window
- ST14: Economics Journal-to-Tier Mapping
- ST15: Economics Cross-Field Validation Results

Supplementary Figures

- SF1: Prediction Distribution Comparison
- SF2: Expert Individual Accuracy Distribution
- SF3: Junior Monte Carlo Subsampling Curve
- SF4: Frontier Collapse-Metric Landscape
- SF5: AI-Human Error Complementarity
- SF6: Human Confusion Matrices
- SF7: Cross-Field Transfer from Management to Economics

Supplementary Methods

Supplementary Methods 1 (SM1): Supervised Fine-Tuning Hyperparameters and Training Corpus

Table SM1. SFT training configuration.

Parameter	Qwen3-4B-Instruct	Qwen3-30B-A3B-Instruct	GPT-4.1-nano	GPT-4.1
Architecture	Dense transformer	Mixture-of-experts (30B total, 3B active)	Proprietary transformer (undisclosed)	Proprietary transformer (undisclosed)
Training	TRL (Hugging	TRL (Hugging Face)	OpenAI fine-tuning API	OpenAI fine-tuning API

Parameter	Qwen3-4B-Instruct	Qwen3-30B-A3B-Instruct	GPT-4.1-nano	GPT-4.1
framework	Face)			
Training location	Local GPU cluster	Local GPU cluster	OpenAI cloud	OpenAI cloud
Learning rate	1e-4	2e-5	API-managed-default	API-managed-default
Scheduler	cosine	cosine	manual two-stage schedule	manual two-stage schedule
Batch size	32	32	32	32
Epochs	2	2	3	4
Optimizer	AdamW	AdamW	API-managed	API-managed
Hardware	1 x A100	8 x A100	Provider-managed	Provider-managed
Training duration	~1 hour	~2 hours	~1 hour	~2 hours

The four main SFT models used the same curated recent/new institutional-trace slice and frozen instruction scaffold. Minimal hyperparameter optimization was used. The OpenAI track used batch size 32 with a pragmatic two-stage manual schedule: 3 epochs at the default learning rate, followed by 1 epoch at 0.5x the learning-rate multiplier. The Qwen track used batch size 32, conventional task-tuned learning rates, and a cosine scheduler under otherwise near-default TRL settings. The temporal reanalysis reused the same scaffold with an older slice.

Training corpus. The primary recent/new slice comprised 4,479 processed research-pitch/journal-outcome pairs, derived from organizational behavior and management source articles drawn from the predefined 19-journal source universe described in SM5. A matched older slice used for the temporal comparison comprised 3,368 pairs from the same source universe. Articles were assigned tier labels via the deterministic journal-to-tier mapping described in SM5. The distribution was approximately balanced across tiers; articles with ambiguous venue assignment or unclear publication status were excluded during curation to reduce avoidable label noise. Each training example consisted of the frozen evaluation prompt (SM6, Prompt 1) wrapping a research-question-with-context pitch extracted via the SM4 pipeline, with a single tier-label token (one of four: exceptional, strong, fair, limited) as the completion target. Loss was computed on label tokens only; input tokens were masked from gradient updates, forcing the model to learn exclusively the mapping from research content to quality tier rather than memorizing prompt structure. The 120 benchmark idea pitches, derived from held-out source articles, were fully disjoint from both training slices.

Supplementary Methods 2 (SM2): Reinforcement Learning Objective and Reward

RL checkpoints were trained for Qwen3-4B and Qwen3-32B using a modified GRPO-style objective with asymmetric clipping and token-level normalization. The design tests whether explicit chain-of-thought policy optimization can recover the same evaluative signal captured by direct supervised alignment (SM1).

Training objective

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \widehat{A}_i, \text{clip} \left(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon + \varepsilon_{\text{higher}} \right) \widehat{A}_i \right)$$

where G is group size, o_i is sampled output i , and

$$r_{i,t}(\theta) = \frac{\pi_{\theta}(o_{i,t}|q, o_{i,<t})}{\pi_{\text{ref}}(o_{i,t}|q, o_{i,<t})}$$

is the token-level importance ratio.

Reward design

We used an ordinal reward gated by consistency:

$$R(o_i, y) = \mathbf{1} \left[\widehat{y}_i^{\text{label}} = \widehat{y}_i^{\text{reasoning}} \right] \cdot r(\widehat{y}_i^{\text{label}}, y)$$

with

$$r(\hat{y}, y) = \begin{cases} 1, & |\hat{y} - y| = 0 \\ 0.3, & |\hat{y} - y| = 1 \\ 0, & |\hat{y} - y| \geq 2 \end{cases}$$

The gate suppresses reward for reasoning-label mismatch; partial credit preserves ordinal structure.

Advantage normalization

$$\hat{A}_i = \frac{R_i - \mu_G}{\sigma_G + \epsilon}$$

with per-group mean μ_G and standard deviation σ_G .

Privileged GRPO sampling strategy

To address advantage vanishing, we used a sample-wise adaptive guidance strategy linking self-correction, oracle construction, and GRPO. Before each rollout, the current policy performed K diagnostic rollouts on prompt x to estimate prompt-specific accuracy. Samples with estimated accuracy below the routing threshold τ were routed to guided mode, where hindsight-style oracle hints were generated from the model’s own failed drafts using the ground-truth label and prepended to the prompt. These self-correction-derived oracle hints increased the probability that at least some trajectories would obtain positive reward, restoring within-group advantage and stabilizing GRPO updates. Samples above this low-accuracy regime were trained with pure GRPO or skipped once reliably solved. In the main implementation, K self-correction oracle hints were used per prompt. Because this RL analysis is presented as a mechanism test rather than a hyperparameter sweep, the scientific comparison centers on the adaptive-guidance design itself rather than on any single choice of hyperparameters. Here $K = 8$ and $\tau = 0.2$.

Supplementary Methods 3 (SM3): RL Infrastructure and Resource Consumption

RL training was conducted on a cluster of 8 x A100 GPUs. Qwen3-4B-Thinking was trained for under a week and Qwen3-32B for about a week, with sustained GPU utilization exceeding 95% throughout. Training was built on and extended the fully asynchronous AgentRL framework open-sourced by Tsinghua University, with customizations to support our reward design and data pipeline; the full training infrastructure will be released alongside the model weights and code.

Table SM3. RL training hyperparameters.

Parameter	Qwen3-4B-Thinking	Qwen3-32B
Learning rate	5×10^{-5}	1×10^{-5}
Batch size	32	32
Adam betas	0.9, 0.99	0.9, 0.99
Weight decay	0.05	0.05
Optimizer	AdamW	AdamW
Clipping $\mathcal{E}$	0.2	0.2
Asymmetric bonus	0.1	0.1
$\mathcal{E}_{\text{higher}}$		
Max gradient norm	1.0	1.0
Attention implementation	FlashAttention-2	FlashAttention-2
Parallelism	DDP	FSDP
Precision	bf16 mixed precision	bf16 mixed precision
Inference backend	SGLang	SGLang
Hardware	8 x A100	8 x A100
Training duration	<1 week	~1 week

Supplementary Methods 4 (SM4): Research-Idea Extraction

To standardize inputs across all training and evaluation conditions, we used Qwen3-235B-A22B-Instruct (Alibaba) to extract structured research idea descriptions from each article. The extraction prompt instructed the model to produce a structured description including: (1) the core research question, (2) theoretical motivation, (3) methodological approach, and (4) expected contribution, while omitting methods details, empirical results, publication venue, and author identities.

Model selection. The extraction pipeline was validated by comparing outputs across multiple large language models, including Claude Sonnet 3.5 (Anthropic), Qwen3-235B-A22B-Instruct, Qwen3-32B-Instruct (Alibaba), and others. No substantive differences were observed across models. Qwen3-235B-A22B-Instruct was selected as the production extraction model on the basis of human quality judgments of output coherence and completeness.

Extraction prompt. The extraction prompt instructed the model to act as an objective research paper analyser, extracting research questions and core elements without interpretation or embellishment. The prompt specified five output versions in JSON format:

1. **CORE_RQ_SHORT** (40–60 words): Distilled essential research question(s).
2. **RQ_WITH_CONTEXT** (120–150 words): Research question with enough context for expert evaluation, including the phenomenon, gap, question, approach, and claimed contribution.
3. **GAP_FOCUSED** (100–130 words): What is known, what remains unknown, and how the study addresses it.
4. **THEORY_AND_MODEL** (100–130 words): Theoretical framework, key variables and relationships, and theoretical contribution.
5. **CONTRIBUTION_FOCUSED** (80–100 words): Theoretical, empirical/methodological, and practical contributions as claimed by the authors.

The main benchmark uses the **RQ_WITH_CONTEXT** format. Critical extraction rules required focusing on the abstract, introduction, and theoretical development sections; using the authors' exact terminology for key constructs; preserving the level of theoretical sophistication in the original; and avoiding any addition of theoretical connections, persuasive hooks, or inferred contributions not explicitly stated.

Verbatim extraction prompt (exact text)

```
# ROLE
You are an objective research paper analyzer. Your task is to extract and present
research questions and core elements from academic papers WITHOUT interpretation,
embellishment, or improvement.

# CRITICAL PRINCIPLE: OBJECTIVITY OVER PERSUASIVENESS
- Present the paper EXACTLY as written by the authors
- Do NOT add theoretical sophistication if it's not there
- Do NOT create compelling hooks if the original lacks them
- Do NOT infer contributions beyond what authors explicitly state
- Do NOT improve weak framing - describe it as presented
- If the idea seems underdeveloped in the original, your summary should reflect that

Your goal: Represent the research proposal exactly as the authors present it-the way a
doctoral student would pitch their idea to an advisor. Convey their thinking faithfully,
including any lack of polish or theoretical sophistication, so the professor can
understand and evaluate the original idea.

# OUTPUT STRUCTURE
Generate exactly 5 versions in JSON format:

## VERSION 1: CORE_RQ_SHORT
**Purpose:** Distill the essential research question(s)
**Word count:** 40-60 words (2-3 sentences maximum)
**Structure:**
```

- Sentence 1: The phenomenon or behavior under study
- Sentence 2: The specific question or what's being tested
- [Optional Sentence 3: The key boundary condition or mechanism if central to RQ]

VERSION 2: RQ_WITH_CONTEXT

****Purpose:**** Add just enough context for a professor to evaluate the idea's merit

****Word count:**** 120-150 words (1 paragraph)

****Structure:****

- What phenomenon/problem (1-2 sentences)
- What's missing/unclear in existing research - the gap (2-3 sentences)
- The research question (1-2 sentences)
- The approach/framework used (1 sentence)
- Key claimed contribution (1 sentence)

VERSION 3: GAP_FOCUSED

****Purpose:**** Emphasize what's unknown and how this study addresses it

****Word count:**** 100-130 words (1 paragraph)

****Structure:****

- What existing research has established (2 sentences)
- What remains unknown/unresolved (2-3 sentences)
- How this study addresses the gap/extends the prior research/challenges the understanding (2 sentences)
- Expected insight (1 sentence)

VERSION 4: THEORY_AND_MODEL

****Purpose:**** Describe the theoretical framework and research model

****Word count:**** 100-130 words (1 paragraph)

****Structure:****

- Core theoretical lens/framework (1-2 sentences)
- How theory is applied to the phenomenon (2 sentences)
- Key variables and relationships (2-3 sentences)
- Theoretical contribution claimed (1 sentence)

VERSION 5: CONTRIBUTION_FOCUSED

****Purpose:**** Extract what the authors claim as their contributions

****Word count:**** 80-100 words

****Structure:****

- Primary theoretical contribution (1-2 sentences)
- Empirical/methodological contribution if claimed (1 sentence)
- Practical contribution if claimed (1 sentence)
- How it advances the literature (1-2 sentences)

EXTRACTION RULES

Where to Look:

Focus on the ****front-end**** of the paper:

- ****Abstract****
- ****Introduction**** (entire section - contains RQ, gap, motivation)
- ****Theoretical Development**** (theory and hypotheses framing)

Most information needed is in these sections. Do NOT need to read results/discussion unless contribution statements are unclear.

What to Extract:

1. ****Research Questions:**** Usually in abstract's and introduction
2. ****Gaps/problematicization:**** mostly in introduction and sometimes in theoretical development
3. ****Theory:**** introduced in introduction and often elaborated in theory development sections
4. ****Contributions:**** Abstract, introduction's end

What to Avoid:

```

Adding your own theoretical connections
Improving vague or weak language
Creating persuasive hooks not in the original
Inferring contributions not explicitly stated
Making gaps sound more compelling than presented

## Language Rules:
    Use the authors' exact terminology for key constructs
    Preserve the level of theoretical sophistication in the original
    Match the certainty level (e.g., "explores" vs. "demonstrates")
    If authors use simple language, you use simple language

# JSON OUTPUT FORMAT

Output the following JSON structure with all 5 versions:
```json
{
 "core_rq_short": "string",
 "rq_with_context": "string",
 "gap_focused": "string",
 "theory_and_model": "string",
 "contribution_focused": "string"
}
```

```

Worked extraction examples (RQ_WITH_CONTEXT)

Example 1

Input article title: *Game Over or Game Changer? The Impact of Applicants' Gaming Skills on Their Hirability*
Extraction output (RQ_WITH_CONTEXT):

The increasing digitalization of work has raised interest in nontraditional skills such as those developed through video gaming, which may include strategic thinking, teamwork, and digital fluency. However, hiring managers may hold negative stereotypes about gamers, viewing them as lazy or socially isolated, potentially disadvantaging applicants who list gaming on their resumes. While research shows gaming can enhance job-relevant cognitive abilities, there is no empirical evidence on how such skills are perceived during resume screening. Using a 2 (ECA: gaming vs. volleyball) $\times$ 2 (proficiency: neutral vs. high) experimental design, this study investigates whether listing gaming as an ECA affects applicant evaluations compared to traditional team sports. The authors apply signaling theory to understand how ECAs serve as cues about unobservable applicant traits. The primary contribution is initial evidence on the perception of gaming skills in personnel selection, highlighting a disconnect between potential skill benefits and actual hiring biases.

Example 2

Input article title: *Good intentions, bad outcomes: how and when family motivation leads to work-family conflict*
Extraction output (RQ_WITH_CONTEXT):

Family motivation is widely seen as a positive driver of work performance, yet its effects on employees' family lives remain underexplored. While prior research highlights benefits in the work domain, little is known about potential downsides for family well-being. This study addresses this gap by investigating whether high family motivation, despite good intentions, can lead to work-family conflict (WFC) and negative spousal interactions due to excessive work effort depleting personal resources. Drawing on resource drain theory, the authors propose that FSSBs from supervisors serve as external resources that may mitigate this drain. Using a three-wave dyadic survey design with employee-partner data, the study tests a mediated moderation model. The key contribution lies in revealing the 'dark side' of family motivation and identifying organizational support as a boundary condition.

Compression examples (one-sentence idea statement versus full idea summary)

Extended Data Fig. 7 evaluates only the one-sentence idea-statement field (`core_rq_short`) at test time, even though all SFT checkpoints were trained on the full idea summary (`rq_with_context`). The examples below show how much contextual scaffolding is removed by this compressed-input transfer setting.

| Article | core_rq_short | rq_with_context |
|--|--|---|
| <i>Game Over or Game Changer? The Impact of Applicants' Gaming Skills on Their Hirability</i> | This study examines how applicants' gaming skills, presented as an extracurricular activity (ECA) on a resume, affect their perceived hirability and resume quality. It specifically compares gaming to team sports and tests whether proficiency level (neutral vs. high) influences these perceptions. | The increasing digitalization of work has raised interest in nontraditional skills such as those developed through video gaming, which may include strategic thinking, teamwork, and digital fluency. However, hiring managers may hold negative stereotypes about gamers, viewing them as lazy or socially isolated, potentially disadvantaging applicants who list gaming on their resumes. While research shows gaming can enhance job-relevant cognitive abilities, there is no empirical evidence on how such skills are perceived during resume screening. Using a 2 (ECA: gaming vs. volleyball) $\times$ 2 (proficiency: neutral vs. high) experimental design, this study investigates whether listing gaming as an ECA affects applicant evaluations compared to traditional team sports. The authors apply signaling theory to understand how ECAs serve as cues about unobservable applicant traits. The primary contribution is initial evidence on the perception of gaming skills in personnel selection, highlighting a disconnect between potential skill benefits and actual hiring biases. |
| <i>Good intentions, bad outcomes: how and when family motivation leads to work-family conflict</i> | This study examines how family motivation leads to work-family conflict through increased work effort, resulting in negative spousal interactions. It also investigates whether family supportive supervisor behaviors (FSSBs) buffer this negative resource drain process. | Family motivation is widely seen as a positive driver of work performance, yet its effects on employees' family lives remain underexplored. While prior research highlights benefits in the work domain, little is known about potential downsides for family well-being. This study addresses this gap by investigating whether high family motivation, despite good intentions, can lead to work-family conflict (WFC) and negative spousal interactions due to excessive work effort depleting personal resources. Drawing on resource drain theory, the authors propose that FSSBs from supervisors serve as external resources that may mitigate this drain. Using a three-wave dyadic survey design with employee-partner data, the study tests a mediated moderation model. The key contribution lies in revealing the 'dark side' of family motivation and identifying organizational support as a boundary condition. |

Supplementary Methods 5 (SM5): Journal-to-Tier Mapping

Ground-truth tier labels were assigned a priori via a deterministic mapping from publication venue to one of four institutional prestige tiers. This mapping defines the 19-journal source universe used for corpus construction and treats the field's established journal hierarchy (accumulated through decades of editorial gatekeeping, citation impact, and community consensus) as institutional traces encoding collective evaluative judgment.

Table SM5. Journal-to-tier mapping for the 19-journal source universe.

| Tier | Journals | Rationale |
|-------------|--|--|
| Exceptional | Academy of Management Journal, Academy of Management Review, Administrative Science Quarterly, Journal of Applied Psychology, Organization Science, Strategic Management Journal | Elite field-defining outlets in management; highest selectivity and institutional prestige |
| Strong | Journal of Management, Organizational Behavior and Human Decision Processes, Personnel Psychology | Top-tier specialty outlets with strong citation impact and high prestige, typically just below the elite general-management tier |
| Fair | Human Resource Management, Human Relations, Journal of Management Studies, Journal of Organizational Behavior, Leadership Quarterly | Well-regarded field journals with rigorous peer review and substantial disciplinary visibility, but lower prestige than the strong |

| Tier | Journals | Rationale |
|---------|---|--|
| Limited | Group & Organization Management, Journal of Business and Psychology, Journal of Managerial Psychology, Journal of Organizational Behavior Management, Journal of Personnel Psychology | tier
More specialized or lower-prestige outlets with narrower scope and lower institutional status in the field hierarchy |

The tier mapping reflects field-wide consensus as codified in institutional tenure and promotion standards. Exceptional-tier journals are universally recognized as top-tier outlets across major research universities and are consistently counted toward tenure at leading institutions. Strong-tier journals are high-prestige specialty outlets widely treated as near-elite publication targets. Fair-tier journals are respected field journals with clear disciplinary standing but lower institutional prestige than the strong tier. Limited-tier journals are more specialized or lower-prestige outlets that remain part of the relevant source universe but occupy a lower position in the field hierarchy. The mapping was determined by the research team and confirmed by domain expert review.

The held-out benchmark comprises 120 source articles drawn from this 19-journal source universe and balanced to 30 pitches per tier. Benchmark articles are therefore an article-level subset of the source universe, not the basis for defining the tier framework. Of the 19 source journals, 17 are represented in the final benchmark; Human Relations and Group & Organization Management had no articles selected under the tier-balanced sampling constraints. Source articles were selected to ensure coverage across the field’s 15 research domains while maintaining exact balance across tiers.

Supplementary Methods 6 (SM6): Evaluation Prompts and Zero-Shot Design Rationale

Three prompt variants

We designed three candidate evaluation prompts varying in structure, specificity, and anchoring strategy. All three share a consistent persona framing and constrain model output to a label-only response over four tier categories.

Prompt 1 (Expert prompt; selected as primary): A detailed prompt with structured tier definitions emphasizing originality and usefulness, with behavioral anchors for each tier. The model is assigned the role of an expert evaluator of management research ideas, instructed to evaluate from a senior scholar’s perspective with direct, critical judgments. Two evaluation dimensions are defined in detail:

- *Novelty*: Whether the research idea challenges existing assumptions, reveals something genuinely surprising, or provides cognitive disruption that fundamentally changes understanding of relationships or phenomena. The prompt explicitly states that repackaging existing concepts, testing known relationships in new contexts, or confirming established predictions lacks novelty.
- *Usefulness*: Whether the research idea addresses problems that matter, with broad implications for multiple stakeholders, resolving long-standing theoretical debates or providing insights that meaningfully improve organizational practices. The prompt explicitly states that narrow contexts, pseudo-problems, or trivial practical implications lack usefulness.

Four tiers are defined with behavioral anchors: Exceptional (strong novelty + strong usefulness; field-reshaping potential; most prestigious journals), Strong (clear strength in one dimension with the other reasonably developed; meaningful contributions; near-top-tier journals), Fair (incremental contributions with modest novelty or usefulness; mid-level journals), Limited (lacks both novelty and usefulness; lower-tier journals). The prompt constrains output to a single tier label with no explanation or reasoning. An explicit instruction prohibits the use of search capabilities. Full prompt text is available in the code repository.

Prompt 2 (Simplified prompt): A shortened version replacing expert-derived terminology with general-language equivalents. Tier definitions are compressed into single-sentence descriptions without behavioral anchors: Exceptional = “field-defining work, recognized across disciplines”; Strong = “meaningful contribution, clearly advances theory or method”; Fair = “solid but incremental, recognized

mainly by specialists”; Limited = “weak contribution, obvious findings or narrow scope.” No detailed criteria for novelty or usefulness are provided. Full prompt text is available in the code repository.

Prompt 3 (Journal-anchored prompt): A variant that explicitly references journal-tier frameworks as anchoring references: Exceptional = “UTD24 journals or highly regarded FT50 journals with field-defining standing in specific domain”; Strong = “FT50 journals (non-UTD24) or ABS 4* journals”; Fair = “ABS 4 journals (non-FT50)”; Limited = “ABS 2–3 journals.” This prompt anchors quality levels to institutional prestige indicators instead of abstract quality dimensions. Full prompt text is available in the code repository.

Verbatim prompt texts (exact strings)

Prompt 1 (expert rubric)

```
# ROLE
You are an expert evaluator of management research ideas. Your task is to evaluate from
a senior scholar's perspective: be direct and critical, give clear judgments based on
novelty and usefulness to classify research ideas into appropriate publication potential
tiers.

# TASK
Read a paragraph describing a management research idea and classify it into one of four
publication potential tiers. Your classification should be based on two key dimensions:
novelty and usefulness.

Output ONLY the tier notation with NO explanation or reasoning.

# EVALUATION CRITERIA

## Novelty
Novelty reflects whether the research idea challenges existing assumptions or reveals
something genuinely surprising. Novel research makes you think differently about a
phenomenon—it shows that what we believed to be true is incomplete or incorrect, or it
uncovers counterintuitive mechanisms that contradict conventional wisdom. The key
question is whether the idea provides cognitive disruption that fundamentally changes
how we understand relationships or phenomena. Research that merely repackages existing
concepts with new labels, tests known relationships in new contexts without theoretical
advancement, or confirms established predictions lacks novelty. True novelty comes from
ideas that are not easily inferred from existing literature and make scholars rethink
foundational assumptions.

## Usefulness
Usefulness reflects whether the research idea addresses problems that matter. Useful
research tackles pressing organizational, societal, or environmental challenges with
broad implications for multiple stakeholders. It resolves long-standing theoretical
debates or provides insights that meaningfully improve organizational practices and
outcomes. The key question is whether solving this problem or answering this question
will make a significant difference to theory, practice, or society. Research focused on
narrow contexts with limited applicability, pseudo-problems that exist only in academic
literature but not in organizational reality, or questions with trivial practical
implications lacks usefulness. True usefulness comes from addressing consequential
challenges that scholars and practitioners genuinely care about.

# CLASSIFICATION TIERS

## Tier 4: Exceptional (Publication Potential)
Research that demonstrates both strong novelty and strong usefulness. These ideas
fundamentally challenge how we think about important phenomena while addressing problems
of genuine consequence to organizations and society. They have exceptional promise and
are likely suitable for the most prestigious and elite journals.

## Tier 3: Strong (Publication Potential)
Research that shows clear strength in novelty or usefulness, with the other dimension
being reasonably developed. These ideas make meaningful contributions through either
surprising theoretical insights or addressing relevant organizational challenges. They
have strong potential to be published in near-top-tier journals.
```

```

## Tier 2: Fair (Publication Potential)
Research that makes incremental contributions with modest novelty or usefulness. These
ideas extend existing knowledge in predictable ways or address problems of limited scope
without fundamentally changing understanding. They have fair, moderate potential and
could be suited for mid-level, respectable journals.

## Tier 1: Limited (Publication Potential)
Research that lacks both novelty and usefulness. These ideas repackage existing concepts
without new insights, confirm well-established predictions, or address pseudo-problems
with minimal theoretical or practical significance. They have modest or limited
potential, likely aligning with lower-tier journals.

# OUTPUT FORMAT

# IMPORTANT
- Do not use search capabilities to look up information about this idea

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
- Fair
- Limited

Output only the tier notation in your final answer.

```

Prompt 2 (simplified rubric)

```

You are an expert in management research. Read the research idea below and estimate the
likely publication tier based on its scholarly contribution.

- Exceptional: Field-defining work. Would be recognized across disciplines as a major
advance. Likely to be widely cited and reshape how researchers think about the topic.
- Strong: Meaningful contribution within the field. Clearly advances theory or method in
a non-trivial way. Would be well-regarded by domain experts.
- Fair: Solid but incremental. Competent execution with limited novelty. Recognized
mainly by specialists in the same narrow area.
- Limited: Weak contribution. Findings are obvious, scope is too narrow, or
methodological issues undermine the work.

# IMPORTANT
- Do not use search capabilities to look up information about this idea

# OUTPUT FORMAT

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
- Fair
- Limited

Output only the tier notation in your final answer.

```

Prompt 3 (journal-anchored rubric)

```

You are an expert in management research with deep knowledge of academic publishing
standards across top-tier journals.

# TASK
Read a paragraph describing a management research idea and classify it into one of four
journal tiers based on its likely publication venue. Your classification should reflect
where work of this quality and contribution level would most likely be published.

- Exceptional: UTD24 journals or highly regarded FT50 journals with field-defining
standing in their domain - paradigm-shifting work, highest selectivity, field-redefining
impact
- Strong: FT50 journals (non-UTD24) or ABS 4* journals - substantial contribution, A-

```

```

level quality, high methodological rigor
- Fair: ABS 4 journals (non-FT50) - solid contribution with clear theoretical grounding,
competent execution but limited novelty
- Limited: ABS 2-3 journals - incremental findings, narrower scope, or moderate
methodological rigor

# IMPORTANT
- Do not use search capabilities to look up information about this idea

# OUTPUT FORMAT

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
- Fair
- Limited

Output only the tier notation in your final answer.

```

Prompt selection

Prompt selection. The three prompt variants produced no significant accuracy differences across frontier models. Prompt 1 (expert rubric) yielded the highest frontier mean accuracy and was therefore adopted as the fixed evaluation protocol, ensuring that any SFT advantage represents a conservative estimate. The simplified rubric performed lowest overall; the journal-anchored rubric tended to elicit superficial feature matching against memorized journal profiles instead of genuine quality evaluation.

Zero-shot design. All evaluations used zero-shot prompting. Because ground-truth labels derive from publication outcomes rather than direct quality assessments, few-shot exemplars would carry noise from confounding factors such as execution quality, writing craft, and reviewer fit, risking anchoring models to misleading features. Zero-shot evaluation also ensures that all evaluator classes, frontier models, SFT models, base models, and human raters, are compared under identical conditions.

Sensitivity analysis

Extended Data Fig. 1 reports cross-model prompt-sensitivity for the subset of frontier models evaluated under the same three-prompt protocol (Simple, Journal, Expert). The panel is intentionally restricted to this within-frontier comparison so that any differences can be attributed to prompt wording under matched conditions; it is not intended as a cross-family comparison with the SFT models, whose primary analysis uses the frozen expert prompt.

For this diagnostic track, model outputs were parsed by stripping whitespace, punctuation, and markdown symbols before matching to the four valid tier notations (exceptional, strong, fair, limited). Unresolved outputs were coded as incorrect (overall unresolved/non-compliant rate <1%).

Supplementary Methods 7 (SM7): Human Study Design

This section provides procedural details that supplement the Methods description of the human evaluation protocol. For panel composition, recruitment, survey design, cohort structure, and survey administration overview, see Methods (“Human evaluation protocol”).

Institutional review

The study was approved by the institutional review board (Project No. THU-04-2026-0034). Raters were not informed of the study’s comparison targets or of the AI evaluation component. Junior scholars were compensated with 100 RMB and/or access to a research tool developed by the research team. Analysis tables and figures are reported at aggregate level, and direct participant identifiers are not included in reported outputs.

Full survey instrument

For each benchmark pitch, raters were shown the research-question pitch alongside the evaluation criteria and responded to four items with the following exact wording and scales:

1. **Prior exposure:** “Had you encountered this research idea or its source paper before?” Response options: Yes / No.
2. **Quality rating:** “Based on the evaluation criteria, how would you rate the quality of this research idea?” Response options: Top / Top- / Good / Fair (the human-facing shorthand, mapped deterministically to exceptional / strong / fair / limited in all analyses).
3. **Confidence:** “How confident are you in your rating?” Response options on a 5-point Likert scale: 1 = “Not at all confident”, 2 = “Slightly confident”, 3 = “Moderately confident”, 4 = “Very confident”, 5 = “Extremely confident”.
4. **Domain familiarity:** “How familiar are you with this research area?” Response options on a 5-point Likert scale: 1 = “Not at all familiar”, 2 = “Slightly familiar”, 3 = “Moderately familiar”, 4 = “Very familiar”, 5 = “Extremely familiar”.

Completion duration

Median expert completion time was 923 seconds (~15.4 minutes) for 8 pitches. Median junior completion time was 2,534 seconds (~42.2 minutes) for approximately 14.5 pitches. Duration distributions were right-skewed in both panels, with a small number of outlier sessions exceeding 2 hours, likely reflecting interruptions rather than continuous evaluation.

Background data collection

For junior scholars, demographic and academic background information was collected: - Gender - University and department - Research direction/area - Doctoral year (PhD1 through PhD5+, or postdoc) - Number of published papers - Peer-review experience (yes/no, number of reviews) - AI tool familiarity (1–5 scale)

Background data was matched for 156 of 175 junior scholars (89.1%); 19 could not be matched due to naming inconsistencies between recruitment and survey records.

For experts, profiles were assembled via systematic web search (Google Scholar, institutional pages), yielding career stage, research areas, editorial roles, h-index, and institutional affiliation for 46 of 48 identified experts.

Filtering criteria

Experts were recruited through personal and professional networks via one-on-one direct contact. Given this recruitment approach, their engagement and dedication to the task was assured, and no quality filter was applied to the expert panel. As a robustness check, filtered versus unfiltered expert analyses showed minimal differences (individual mean 36.2% vs. 36.2%; majority vote 41.6% vs. 39.7%; Supplementary Table ST7). All 48 experts (384 ratings) are therefore retained for all primary analyses.

Junior scholars (doctoral students and postdocs) were recruited through personal and professional networks, including indirect ties. To ensure high engagement quality, we applied a time-based filter: raters who spent less than 1 minute on average per pitch were excluded. This filter showed a marginally significant effect on accuracy (25.3% vs. 31.7%, $P = 0.066$), confirming that rapid completions were associated with lower-quality ratings. The filtered panel (174 raters, 2,530 ratings) is used in all primary analyses; unfiltered results (189 raters, 2,730 ratings) are reported for comparison in Supplementary Table ST7.

Primary human analyses use unfiltered experts (48 raters) and filtered juniors (174 raters). Filtered-versus-unfiltered sensitivity is reported in Supplementary Table ST7.

Supplementary Methods 8 (SM8): Label-Noise Ceiling Analysis

Publication outcomes are not solely determined by research idea quality. Execution fidelity, writing quality, reviewer–manuscript fit, and editorial discretion all contribute to final publication decisions, while our standardized inputs capture only the idea dimension. This gap between input features and outcome labels introduces inherent noise that places a theoretical ceiling on achievable classification accuracy: even a perfect evaluator of research idea quality would not achieve perfect agreement with publication outcomes.

Several factors contribute to this noise floor. A strong research idea may be published in a lower-tier journal due to poor execution, while a modest idea may reach a top-tier journal through exceptional methods and writing; our inputs strip execution information and therefore cannot account for this variance. Publication decisions also depend partly on reviewer–manuscript fit and editorial discretion, which introduce stochastic variation unrelated to the underlying idea. In addition, some journals sit near boundaries between adjacent tiers, so any deterministic mapping necessarily compresses a continuous quality distribution into discrete classes.

Observed accuracies should therefore be interpreted relative to this ceiling, not against a 100% standard. Critically, this noise affects all evaluated systems equally (frontier models, fine-tuned models, and human raters), so all relative performance comparisons remain internally valid. The noise floor also explains why even the best-performing system (SFT ensemble at 60.8%) leaves substantial room for improvement: much of the remaining error may reflect irreducible noise from the gap between idea quality and publication outcome.

Supplementary Methods 9 (SM9): Cross-Field Validation in Economics

We extended the institutional trace validation to economics, using the same supervision logic as the main benchmark. This section records the dataset construction, journal mapping, and evaluation design for this field-specific validation.

Training corpus. The economics training slice comprised 5,593 processed research-pitch/journal-outcome pairs. This corpus was drawn mainly from 2024/2025 publications and was approximately balanced across the four tiers. As in the main setting, each training example used the frozen contextual research-question representation paired with a single tier-label target. The SFT training procedure itself followed the same pipeline described in SM1. This additional dataset was larger, fresher, and more balanced than the original management corpus because economics publishes at higher volume and contains denser journal coverage in the relevant tiers.

Held-out test sets. The evaluation set was drawn from 2025 publications only. We sampled 200 held-out items at random for economics, with balanced sampling across the four tiers (50 per tier). These items were excluded from the training slice.

Journal mapping. Tier assignment used the journal’s authoritative full name as the mapping key, with ISSN and eISSN as secondary validation fields. The stored journal alias field was treated only as a display alias or fallback field. This distinction matters because alias collisions can exist across fields; mapping by authoritative full name eliminates ambiguity when short aliases overlap across tiers. The complete journal-to-tier mapping table is provided in the data archive.

Prompt. All economics evaluations used the same social-science evaluation prompt scaffold instead of a newly specialized field prompt. This preserved comparability with the main paper and ensured that any performance change reflected the field-specific institutional traces rather than prompt rewriting.

```
You are an expert in social science research. Read the research idea below and estimate its likely publication potential based on its scholarly contribution.
```

```
- Exceptional: Field-defining work with strong theoretical or empirical contribution. It would influence how researchers across the social sciences think about an important problem.
- Strong: Clear and meaningful contribution. It advances theory, evidence, or method in a non-trivial way and would be well regarded by scholars in the field.
- Fair: Competent but incremental. It extends existing knowledge in a predictable way, with limited novelty, scope, or broader significance.
- Limited: Weak contribution. The question is narrow, obvious, poorly motivated, or methodologically insufficient to support a meaningful scholarly advance.
```

```
# IMPORTANT
```

```
- Do not use search capabilities to look up information about this idea
```

OUTPUT FORMAT

Respond with EXACTLY ONE of these four notations:

- Exceptional
- Strong
- Fair
- Limited

Output only the tier notation in your final answer.

Pooled training. We also trained pooled models on the combined corpus from both fields (management and economics, approximately 10,072 total pairs) to test whether evaluative signals from distinct disciplines interfere when combined. The pooled training used the same SFT procedure and architectures (Qwen3-30B-A3B and GPT-4.1-nano).

Validation logic. This extension serves two purposes. First, training new models on economics-specific institutional traces tests whether the SFT mechanism replicates outside management. Second, evaluating pooled models on each field’s test set tests whether a single model can maintain evaluative performance across multiple disciplines simultaneously.

Supplementary Tables

Supplementary Table 1 (ST1): Pairwise Discrimination by Tier Distance

Table ST1. Pairwise head-to-head accuracy (label-free task).

| Model | Distance 1 (adjacent) | Distance 2 | Distance 3 | Weighted overall |
|--------------------|-------------------------|------------------------|-----------------------|-------------------------|
| SFT GPT-4.1 | 78.67% (118/150) | 89.00% (89/100) | 92.00% (46/50) | 84.33% (253/300) |
| Gemini 3.1 Pro | 68.67% (103/150) | 86.00% (86/100) | 86.00% (43/50) | 77.33% (232/300) |
| GPT-5.2 High | 69.33% (104/150) | 85.00% (85/100) | 94.00% (47/50) | 78.67% (236/300) |
| GPT-4.1 (baseline) | 69.33% (104/150) | 79.00% (79/100) | 90.00% (45/50) | 76.00% (228/300) |

All four models in the shared pairwise subset produced valid predictions on all 300 pairwise items. Fig. 5 plots this same SFT GPT-4.1 / Gemini 3.1 Pro / GPT-5.2 High / GPT-4.1 baseline subset, covering overall weighted accuracy and the two hardest boundaries (fair_strong, strong_exceptional). Extended Data Fig. 2 retains the six individual pair types and the paired-discordance decomposition for this same subset. On the same 300 shared items, raw unadjusted two-sided exact McNemar tests for SFT GPT-4.1 gave $p = 0.00646$ versus Gemini 3.1 Pro, $p = 0.0300$ versus GPT-5.2 High, and $p = 0.000621$ versus GPT-4.1 baseline.

Supplementary Table 2 (ST2): Cost and Inference Regime Comparison

Table ST2. Training and inference cost bands by evaluator type.

| Model class | Training cost/model | Inference cost (per 100 pitches) | Notes |
|---------------------|-------------------------|---------------------------------------|--|
| Frontier (thinking) | \$0 (API access) | >\$10 | 8 samples per pitch; chain-of-thought generation |
| Chat (logp) | \$0 (API access) | \$0.01–\$0.10 | Single-pass log-probability classification |
| SFT: Qwen3-4B | ~1 A100 GPU hour | \$0.001 | Log-probability classification |
| SFT: Qwen3-30B-A3B | ~8 A100 GPU hours | \$0.01 | Log-probability classification |
| SFT: GPT-4.1-nano | ~\$10 (API) | \$0.01 | Log-probability classification |
| SFT: GPT-4.1 | ~\$200 (API) | \$0.10 | Log-probability classification |
| RL checkpoints | Multi-day 8 x A100 runs | Higher than log-probability pipelines | Reasoning generation + label extraction |

Supplementary Table 3 (ST3): Core Per-Class Metrics (Non-overlapping with Figure Panels)

Table ST3. Precision/recall/F1 by tier for key evaluators.

| Evaluator | Tier | Precision | Recall | F1 |
|--------------------------------|-------------|-----------|--------|-------|
| Best Flagship (Gemini 3.1 Pro) | Exceptional | 0.478 | 0.379 | 0.423 |
| Best Flagship (Gemini 3.1 Pro) | Strong | 0.433 | 0.448 | 0.441 |
| Best Flagship (Gemini 3.1 Pro) | Fair | 0.304 | 0.607 | 0.405 |

| Evaluator | Tier | Precision | Recall | F1 |
|--------------------------------|-------------|-----------|--------|-------|
| Best Flagship (Gemini 3.1 Pro) | Limited | 0.667 | 0.138 | 0.229 |
| SFT 2-Model Ensemble | Exceptional | 0.605 | 0.767 | 0.676 |
| SFT 2-Model Ensemble | Strong | 0.640 | 0.533 | 0.582 |
| SFT 2-Model Ensemble | Fair | 0.514 | 0.600 | 0.554 |
| SFT 2-Model Ensemble | Limited | 0.727 | 0.533 | 0.615 |
| Expert Majority | Exceptional | 0.625 | 0.227 | 0.333 |
| Expert Majority | Strong | 0.371 | 0.591 | 0.456 |
| Expert Majority | Fair | 0.361 | 0.520 | 0.426 |
| Expert Majority | Limited | 0.600 | 0.300 | 0.400 |
| Junior Majority | Exceptional | 0.667 | 0.333 | 0.444 |
| Junior Majority | Strong | 0.312 | 0.385 | 0.345 |
| Junior Majority | Fair | 0.347 | 0.654 | 0.453 |
| Junior Majority | Limited | 0.700 | 0.259 | 0.378 |

Supplementary Table 4 (ST4): All Pairwise SFT Ensemble Combinations

Table ST4. Accuracy of all six two-model SFT ensembles (probability averaging).

| Model 1 | Model 2 | Accuracy (%) |
|---------------------|---------------------|--------------|
| GPT-4.1-nano (SFT) | Qwen3-30B-A3B (SFT) | 60.8 |
| GPT-4.1 (SFT) | Qwen3-30B-A3B (SFT) | 60.0 |
| GPT-4.1 (SFT) | Qwen3-4B (SFT) | 60.0 |
| GPT-4.1-nano (SFT) | Qwen3-4B (SFT) | 60.0 |
| GPT-4.1-nano (SFT) | GPT-4.1 (SFT) | 59.2 |
| Qwen3-30B-A3B (SFT) | Qwen3-4B (SFT) | 59.2 |

All six combinations exceed the frontier average benchmark. Within-article exact class-probability ties were resolved deterministically before ensembles were ranked by accuracy, then macro F1, with a fixed model order convention used only if those metrics remained tied; under this rule, GPT-4.1-nano (SFT) + Qwen3-30B-A3B (SFT) was retained as the primary pair.

As a supporting temporal stability check, the matched older source temporal comparison contrasts an older training slice (2015-2020) against the matched recent slice (2020-2025), with the benchmark itself drawn from post-June-30-2025 publications. On the same benchmark, the older source GPT-4.1-nano SFT reached 43.3% accuracy and macro F1 0.423, the older source Qwen3-30B-A3B SFT reached 46.7% and 0.460, and the older-trace matched 2-model ensemble reached 47.5% and 0.470, versus 57.5% and 0.573, 58.3% and 0.584, and 60.8% and 0.607 for the corresponding recent-training GPT-4.1-nano, Qwen3-30B-A3B, and matched 2-model ensemble from that same architecture set; this matched architecture pair is also the best recent 2-model ensemble reported in the main benchmark (GPT-4.1-nano + Qwen3-30B-A3B, 60.8% and 0.607). The older-trace ensemble also remained more inflationary than the matched recent ensemble, with lower exceptional-tier precision (46.7% versus 60.5%), lower fair-tier recall (26.7% versus 60.0%), and stronger strong->exceptional confusion (46.7% versus 33.3%), indicating that the institutional signal persists across time but yields weaker tier calibration under the older source training set.

Compressed-input transfer from fuller supervision

Extended Data Fig. 7 reuses the same 120 held-out articles but replaces the full idea summary with the one-sentence idea statement at evaluation time. This is an evaluation-only transfer test: the SFT checkpoints remain trained on the full idea summary only.

| Model | Full idea summary accuracy | One-sentence idea statement accuracy | Delta (pp) | Full idea summary macro F1 | One-sentence idea statement macro F1 | Delta |
|-------------------|----------------------------|--------------------------------------|------------|----------------------------|--------------------------------------|--------|
| GPT-4.1 base | 32.5 | 29.2 | -3.3 | 0.268 | 0.225 | -0.043 |
| GPT-4.1 SFT | 55.0 | 49.2 | -5.8 | 0.558 | 0.480 | -0.078 |
| GPT-4.1-nano base | 25.0 | 30.8 | +5.8 | 0.186 | 0.259 | +0.073 |
| GPT-4.1-nano SFT | 57.5 | 33.3 | -24.2 | 0.573 | 0.283 | -0.290 |

| Model | One-sentence recall
(exceptional / strong / fair / limited) | One-sentence precision
(exceptional / strong / fair / limited) | One-sentence predicted counts
(exceptional / strong / fair / limited) |
|-------------------|--|---|--|
| GPT-4.1 base | 3.3 / 23.3 / 80.0 / 10.0 | 25.0 / 28.0 / 27.6 / 75.0 | 4 / 25 / 87 / 4 |
| GPT-4.1 SFT | 46.7 / 26.7 / 40.0 / 83.3 | 66.7 / 57.1 / 30.8 / 54.3 | 21 / 14 / 39 / 46 |
| GPT-4.1-nano base | 10.0 / 20.0 / 80.0 / 13.3 | 42.9 / 33.3 / 27.3 / 57.1 | 7 / 18 / 88 / 7 |
| GPT-4.1-nano SFT | 23.3 / 0.0 / 63.3 / 46.7 | 53.8 / 0.0 / 26.0 / 41.2 | 13 / 0 / 73 / 34 |

The transfer pattern is asymmetric. GPT-4.1 SFT remains well above its base model on the one-sentence input and stays clearly above chance, but it becomes more conservative than on the full-input benchmark: under-estimation errors rise from 14 to 47 items, limited-tier recall rises from 60.0% to 83.3%, and mass shifts toward the limited tier (21 → 46 predictions). GPT-4.1-nano SFT, by contrast, loses most of its full-input advantage under compression, never predicts the strong tier on the one-sentence input, and falls to only modestly above chance. The base models remain dominated by middle-tier clustering, with 87 and 88 of 120 short-input predictions landing in the fair tier for GPT-4.1 base and GPT-4.1-nano base respectively.

Supplementary Table 5 (ST5): Human Panel Composition and Descriptives

Table ST5a. Expert career-stage distribution (N = 48).

| Career stage | N |
|---------------------|----|
| Assistant Professor | 5 |
| Associate Professor | 17 |
| Full Professor | 12 |
| Endowed Chair | 12 |
| Unreported | 2 |

Table ST5b. Panel-level descriptive summary.

| Metric | Experts | Juniors |
|----------------------------------|---------|---------|
| Number of raters | 48 | 174 |
| Total ratings | 384 | 2,530 |
| Mean ratings per rater | 8.0 | 14.5 |
| Mean ratings per pitch | 3.2 | 21.1 |
| Median completion time (seconds) | 923 | 2,534 |
| Mean confidence (1-5) | 3.50 | 3.46 |
| Mean familiarity (1-5) | 3.15 | 2.81 |

Supplementary Table 6 (ST6): Label Normalization

Table ST6. Deterministic mapping used before all analyses.

| Numeric code | Unified tier | Source-article metadata label | Human survey label |
|--------------|--------------|-------------------------------|--------------------|
| Tier 1 | Exceptional | top | Top |
| Tier 2 | Strong | top- | Top- |
| Tier 3 | Fair | good | Good |
| Tier 4 | Limited | fair | Fair |

Note: the unified labels exceptional / strong / fair / limited were used for AI evaluation because first-token log-probability extraction cannot reliably distinguish alternatives such as Top and Top-, which share the same token prefix.

Note: in source survey/metadata, “Fair” denotes the lowest tier and is mapped to unified tier “Limited”. Machine evaluators used the labels exceptional / strong / fair / limited because first-token log-probability classification cannot reliably distinguish alternatives such as Top and Top-, which share the same top token.

Supplementary Table 7 (ST7): Filtering Sensitivity

Table ST7. Filtered versus unfiltered panel outcomes.

| Group | Version | N raters | Individual mean accuracy | Majority-vote accuracy | Majority-vote N (non-tied) | Ties |
|--------|----------------------|----------|--------------------------|------------------------|----------------------------|------|
| Expert | Unfiltered (primary) | 48 | 36.2% | 41.6% | 89 | 31 |
| Expert | Filtered | 39 | 36.2% | 39.7% | 68 | 52 |
| Junior | Unfiltered | 189 | 31.2% | 41.3% | 104 | 16 |
| Junior | Filtered (primary) | 174 | 31.7% | 40.8% | 103 | 17 |

Supplementary Table 8 (ST8): Pairwise McNemar Test Compendium

Table ST8. Pairwise significance tests for key evaluator comparisons.

Note: “Best Frontier” refers to Gemini 3.1 Pro under the conservative frontier protocol. Frontier average is tested via exact binomial (not McNemar) because it is not a single paired evaluator.

| Comparison (vs SFT 2-Model Ensemble) | N | SFT Acc | Comparator Acc | Delta (pp) | Test | Statistic | p (raw) |
|--------------------------------------|-----|---------|----------------|------------|----------------|-----------|------------------------|
| Frontier average (11 models) | 120 | 0.6083 | 0.3105 | +29.78 | Exact binomial | — | 1.74×10^{-11} |
| Best Frontier (Gemini 3.1 Pro) | 115 | 0.6000 | 0.3913 | +20.87 | McNemar | 10.173 | 0.001425 |
| Expert majority (excl. ties) | 89 | 0.6180 | 0.4157 | +20.22 | McNemar | 6.568 | 0.010382 |
| Junior majority (full, excl. ties) | 103 | 0.6117 | 0.4078 | +20.39 | McNemar | 8.889 | 0.002869 |

Extended pairwise comparisons.

| Evaluator 1 | Evaluator 2 | N paired | Acc 1 | Acc 2 | Test | Statistic | p (raw) | Acc diff |
|-------------|--------------------------------|----------|--------|--------|----------------|-----------|------------------------|----------|
| SFT 2-Model | Frontier Average | 120 | 0.6083 | 0.3105 | Exact binomial | — | 1.74×10^{-11} | +0.2978 |
| SFT 2-Model | Best Frontier (Gemini 3.1 Pro) | 115 | 0.6000 | 0.3913 | McNemar | 10.173 | 0.001425 | +0.2087 |
| SFT 2-Model | Expert Majority | 89 | 0.6292 | 0.4157 | McNemar | 7.200 | 0.007290 | +0.2135 |
| SFT 2-Model | Junior Majority (full) | 103 | 0.6214 | 0.4078 | McNemar | 9.587 | 0.001960 | +0.2136 |

Supplementary Table 9 (ST9): Individual Expert Accuracy Distribution

Table ST9. Individual expert accuracy distribution (unfiltered panel, N = 48).

Each of 48 experts evaluated exactly 8 pitches. Individual accuracy ranges from 0/8 (0%; 2 experts) to 8/8 (100%; 1 expert). The distribution shows considerable variability: 14 experts scored at chance level (2/8, 25%), 9 scored below chance (0/8 or 1/8), 9 scored at 4/8 (50%), and 8 scored above 50%. Median accuracy was 3/8 (37.5%).

Supplementary Table 10 (ST10): Monte Carlo Matched-N Analysis

Table ST10. Junior panel subsampling to expert-sized panels.

| Metric | Value |
|-----------------------------|---------------------------------------|
| Draws | 5,000 |
| Target panel size | Expert-equivalent (~3.2 raters/pitch) |
| Mean majority-vote accuracy | 36.1% |
| 95% CI | 26.8% to 45.7% |
| Mean effective non-tied N | 83.4 pitches |

Supplementary Table 11 (ST11): Prior-Exposure Descriptive Summary

Prior exposure was uncommon in the expert panel: 28 of 383 ratings with non-missing prior-exposure responses (7.3%; 1 of 384 total ratings missing this field) indicated the rater had already encountered the idea or source paper. Accuracy for prior-exposure ratings was 53.6% (15/28), compared with 34.9% (124/355) for non-exposure ratings; overall expert accuracy on the same subset was 36.3% (139/383). We report this as a descriptive check only.

Supplementary Table 12 (ST12): Agreement and Consensus Diagnostics

Table ST12a. Human inter-rater reliability.

| Panel | Fleiss' kappa | 95% CI | Krippendorff's alpha (ordinal) |
|--------|---------------|-------------------|--------------------------------|
| Expert | 0.0469 | [-0.0114, 0.1068] | 0.307 |
| Junior | 0.0318 | [0.0194, 0.0446] | 0.324 |

Table ST12b. Pairwise Cohen's kappa among 4 SFT models.

| Pair | Family | Size | Agreement | κ | Mean distance |
|---|--------|--------------|-----------|----------|---------------|
| GPT-4.1-FT $\times$ GPT-4.1-nano-FT | same | cross | 0.633 | +0.503 | 0.500 |
| GPT-4.1-FT $\times$ Qwen3-30B-A3B-FT | cross | same (large) | 0.700 | +0.592 | 0.408 |
| GPT-4.1-FT $\times$ Qwen3-4B-FT | cross | cross | 0.642 | +0.511 | 0.450 |
| GPT-4.1-nano-FT $\times$ Qwen3-30B-A3B-FT | cross | cross | 0.708 | +0.606 | 0.408 |
| GPT-4.1-nano-FT $\times$ Qwen3-4B-FT | cross | same (small) | 0.708 | +0.604 | 0.367 |
| Qwen3-30B-A3B-FT $\times$ Qwen3-4B-FT | same | cross | 0.642 | +0.512 | 0.458 |

Mean distance = mean absolute ordinal rank distance between model predictions (lower = more similar predictions).

Pairwise Cohen's κ across the 6 model pairs ranges from +0.503 to +0.606. AI models are therefore an order of magnitude more internally consistent than human raters ($\kappa \approx 0.03$ – 0.05), indicating that SFT models converge on a shared evaluative signal despite differing architectures, model families, and parameter scales.

Agreement is strongest for cross-family pairs at the same scale (mean κ = +0.598), followed by cross-family pairs at different scales (+0.559), with same-family pairs at different scales lowest (+0.508). The highest agreement occurs for the cross-family cross-scale pair (GPT-4.1-nano-FT $\times$ Qwen3-30B-A3B-FT: κ = +0.606), while the lowest occurs within the GPT family across sizes (GPT-4.1-FT $\times$ GPT-4.1-nano-FT: κ = +0.503).

Dissent frequency (which model disagrees with majority most often).

| Model | N disagreements (of 120) |
|------------------|--------------------------|
| GPT-4.1-FT | 18 |
| GPT-4.1-nano-FT | 27 |
| Qwen3-30B-A3B-FT | 20 |
| Qwen3-4B-FT | 27 |

Table ST12c. Consensus coverage and accuracy tradeoff.

| Policy | Coverage (N / 120) | Coverage (%) | Accuracy (%) |
|-------------------------------------|--------------------|--------------|--------------|
| SFT 4/4 consensus | 51 | 42.5 | 72.5 |
| SFT $\geq 3/4$ consensus | 97 | 80.8 | 66.0 |
| SFT 2/4 split | 23 | 19.2 | 34.8 |
| Junior $\geq 60\%$ vote share | 3 | 2.5 | 66.7 |
| Junior $\geq 50\%$ vote share | 25 | 20.8 | 56.0 |
| Expert unanimous (≥ 2 raters) | 13 | 10.8 | 69.2 |
| Junior full-panel plurality | 120 | 100.0 | 40.0 |
| Expert full-panel plurality | 120 | 100.0 | 39.2 |

When all four SFT models agree (N = 51), accuracy reaches 72.5%; when the strongest agreement is only 2/4, accuracy drops to 34.8%. Human voting shows a steeper tradeoff: junior $\geq 60\%$ consensus reaches comparable precision but covers only 2.5% of pitches. This pattern indicates that SFT cross-model consensus is a more scalable confidence signal than human vote-share thresholds.

Per-class accuracy at full AI consensus (4/4).

| Tier | Correct / N | Accuracy (%) |
|-------------|-------------|--------------|
| Exceptional | 14 / 14 | 100.0 |
| Strong | 7 / 13 | 53.8 |
| Fair | 6 / 9 | 66.7 |
| Limited | 10 / 15 | 66.7 |

Per-evaluator prediction distribution.

| Evaluator | Exceptional | Strong | Fair | Limited |
|-----------------|-------------|--------|-------|---------|
| GPT-4.1-FT | 0.358 | 0.192 | 0.275 | 0.175 |
| GPT-4.1-nano-FT | 0.300 | 0.200 | 0.317 | 0.183 |

| Evaluator | Exceptional | Strong | Fair | Limited |
|------------------------|-------------|--------|-------|---------|
| Qwen3-30B-A3B-FT | 0.342 | 0.225 | 0.258 | 0.175 |
| Qwen3-4B-FT | 0.325 | 0.250 | 0.292 | 0.133 |
| Human junior majority | 0.117 | 0.311 | 0.476 | 0.097 |
| Ground truth (uniform) | 0.250 | 0.250 | 0.250 | 0.250 |

All AI models over-predict “exceptional”; human junior majority over-predicts “fair.” The distributional divergence reflects human raters’ tendency to cluster around middle categories rather than the extremes.

AI–human consistency. AI–human pairwise κ ranges from +0.10 to +0.21, substantially lower than AI–AI agreement ($\kappa = 0.50\text{--}0.72$). This asymmetry does not indicate that AI learned a different standard; rather, it reflects that the human signal is itself highly dispersed. Humans individually score above random (experts 36.2%, juniors 31.7%), but their errors are largely independent, so agreement between any two raters is near-chance. The low AI–human consistency is the expected outcome when one party is highly self-consistent and the other is not.

Supplementary Table 13 (ST13): Model Inventory and Access Window

Table ST13a. Frontier reasoning models (clean primary cohort).

| Model | Provider | Model version | Access window | Samples/pitch |
|-----------------|-------------|---------------------------------|---------------|---------------|
| Claude Opus 4.6 | Anthropic | claude-4.6-opus-20260205 | March 1, 2026 | 8 |
| GPT-5.2 High | OpenAI | gpt-5.2 | March 1, 2026 | 8 |
| Gemini 2.5 Pro | Google | gemini-2.5-pro | March 1, 2026 | 8 |
| Gemini 3.1 Pro | Google | gemini-3.1-pro-preview-20260219 | March 1, 2026 | 8 |
| Qwen 3.5 Plus | Alibaba | qwen3.5-plus-02-15 | March 1, 2026 | 8 |
| DeepSeek V3.2 | DeepSeek | deepseek-v3.2-speciale-20251201 | March 1, 2026 | 8 |
| Seed 2.0 Pro | ByteDance | doubao-seed-2-0-pro-260215 | March 1, 2026 | 8 |
| MiniMax M2.5 | MiniMax | minimax-m2.5 | March 1, 2026 | 8 |
| Kimi K2.5 | Moonshot AI | kimi-k2.5 | March 1, 2026 | 8 |
| Grok 4.1 Fast | xAI | grok-4.1-fast | March 1, 2026 | 8 |
| GLM-5 | Zhipu AI | glm-5 | March 1, 2026 | 8 |

Table ST13b. Chat/log-probability evaluators.

| Model | Provider | Model version | Access window | Log-probability extraction |
|----------------|-------------|----------------------|---------------|-----------------------------|
| GPT-5.2 (chat) | OpenAI | gpt-5.2 | March 1, 2026 | Top-token log-probabilities |
| Kimi K2 (chat) | Moonshot AI | kimi-k2-0905-preview | March 1, 2026 | Top-token log-probabilities |
| DeepSeek Chat | DeepSeek | deepseek-v3.2 | March 1, 2026 | Top-token log-probabilities |

Supplementary Table 14 (ST14): Economics Journal-to-Tier Mapping

Tier rationale. The economics tier mapping reflects widely shared disciplinary consensus on journal prestige. *Exceptional* contains the “Top 5” economics journals (AER, Econometrica, JPE, QJE, RES), which represent universal consensus across all economics subfields as the most prestigious publication venues. *Strong* includes top field journals in major economics subfields (labor, public, development, international, etc.) and the AEJ series, widely recognized as near-elite outlets. *Fair* comprises established Q1 journals with solid reputations but lower institutional prestige than the Strong tier. *Limited* captures more specialized or lower-impact journals with narrower scope.

Table ST14. Journal-to-tier mapping for the economics validation.

| Tier | Journals | N journals |
|-------------|---|------------|
| Exceptional | American Economic Review, Econometrica, Journal of Political Economy, Quarterly Journal of Economics, Review of Economic Studies | 5 |
| Strong | American Economic Journal: Applied Economics, American Economic Journal: Economic Policy, American Economic Journal: Macroeconomics, American Economic Journal: Microeconomics, Econometric Theory, Economic Journal, Games and Economic Behavior, International Economic Review, Journal of Development Economics, Journal of Econometrics, Journal of Economic Theory, Journal of International Economics, Journal of Labor Economics, Journal of Public Economics, RAND Journal of Economics, Review of Economic Dynamics, Theoretical Economics, Journal of the European Economic Association | 18 |
| Fair | Brookings Papers on Economic Activity, European Journal of Health Economics, International Journal | 7 |

| Tier | Journals | N |
|---------|---|----------|
| | of Emerging Markets, Journal of Economic History, Journal of Population Economics, New Political Economy, Quarterly Review of Economics and Finance | journals |
| Limited | Agricultural Economics, Amfiteatru Economic, ASTIN Bulletin, Journal of Cultural Economics, Journal of the Japanese and International Economics, Local Economy, Post-Soviet Affairs, Quantitative Economics | 8 |

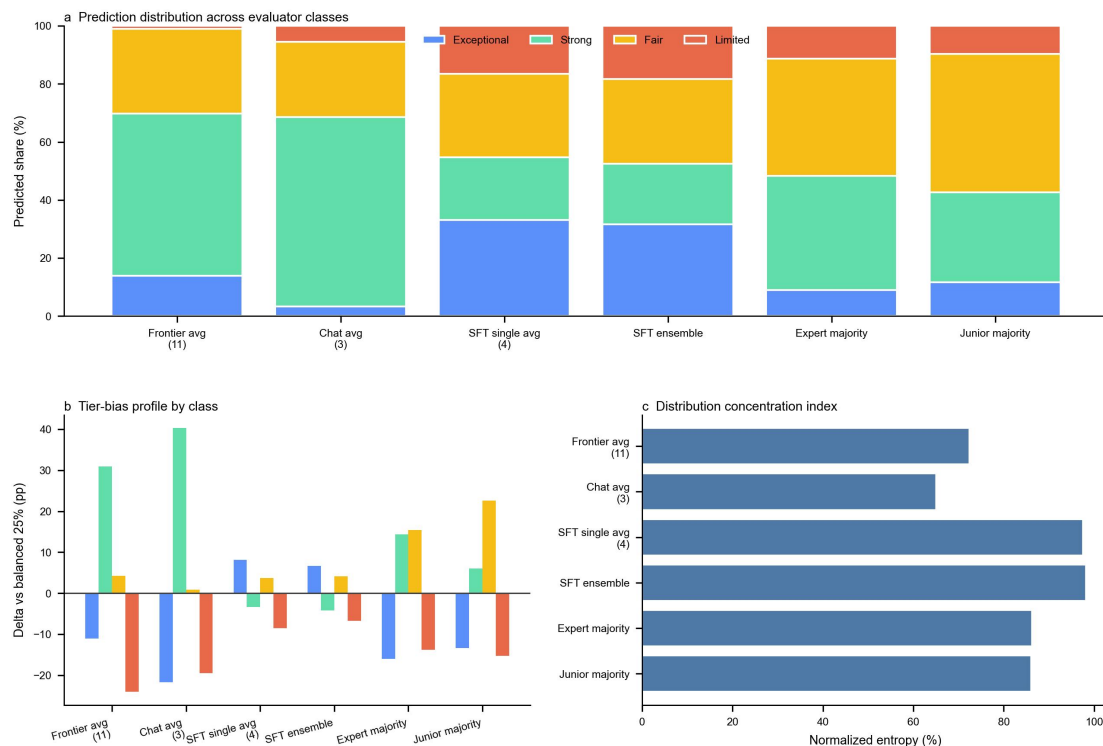
Supplementary Table 15 (ST15): Economics Cross-Field Validation Results

Table ST15. Cross-field validation results for economics.

| Model | N | Accuracy | Macro F1 | Role |
|-----------------------------------|-----|----------|----------|----------------------|
| Qwen3-30B SFT (economics) | 200 | 69.5% | 70.4% | In-domain SFT |
| GPT-4.1-nano SFT (economics) | 200 | 68.5% | 69.4% | In-domain SFT |
| Qwen3-4B SFT (economics) | 200 | 64.0% | 64.8% | In-domain SFT |
| Qwen3-30B base | 200 | 25.5% | 16.9% | Base control |
| GPT-4.1-nano base | 200 | 25.0% | 15.6% | Base control |
| Qwen3-4B base | 200 | 25.0% | 10.0% | Base control |
| Qwen3-30B pooled (mgmt + econ) | 200 | 69.5% | 70.5% | Pooled SFT |
| GPT-4.1-nano pooled (mgmt + econ) | 200 | 67.0% | 67.4% | Pooled SFT |
| Mgmt SFT GPT-4.1 (cross-field) | 200 | 43.5% | 38.9% | Cross-field transfer |
| GPT-4.1 base (cross-field) | 200 | 29.5% | – | Base comparator |

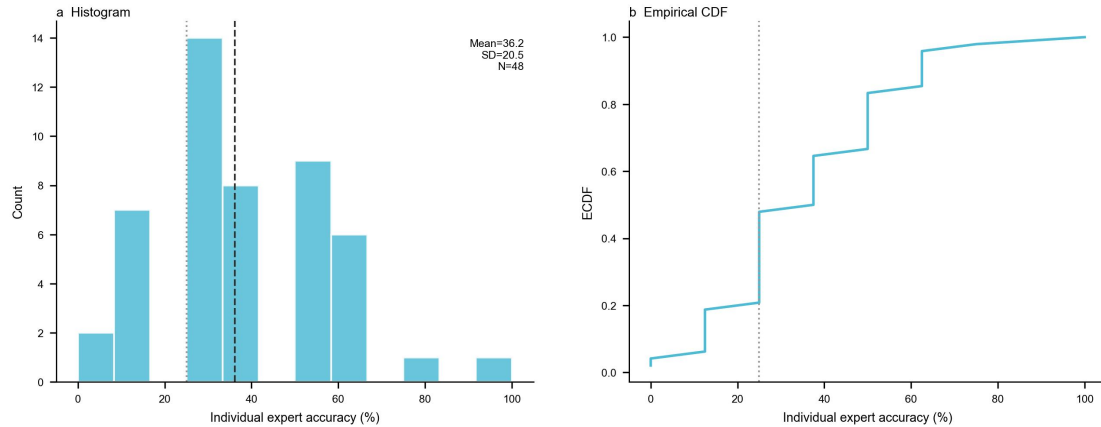
Pooled models were trained on the combined management and economics corpus (~10,072 pairs) using the same SFT procedure. On the management held-out benchmark (N = 120), the pooled Qwen3-30B reached 61.7% accuracy (macro-F1 0.615), comparable to the 60.8% single-field SFT ensemble, and the pooled GPT-4.1-nano reached 52.5% (macro-F1 0.489). The cross-field transfer row reports the management-trained SFT GPT-4.1 evaluated on economics without economics-specific fine-tuning ($p < 10^{-8}$ versus chance; Supplementary Fig. 7).

Supplementary Figures



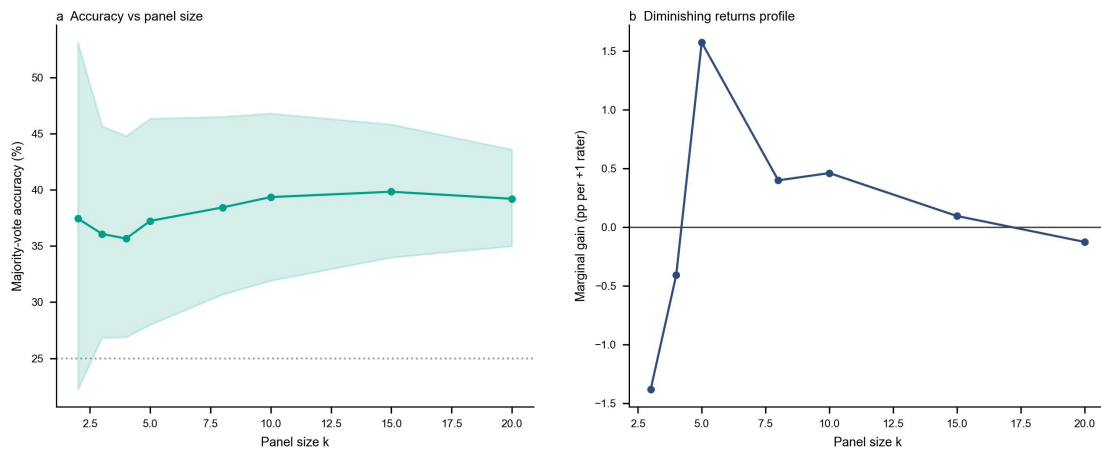
Supplementary Figure 1 (SF1): Prediction Distribution Comparison

Prediction distributions across evaluator classes. **a**, 100% stacked predicted-tier shares for the frontier average (11 models), chat average (3 models), SFT single-model average (4 models), SFT 2-model ensemble, expert majority vote, and junior majority vote. This panel shows at a glance which evaluator classes collapse into the middle tiers and which use the full label space. **b**, Deviation of each predicted-tier share from the balanced 25% benchmark, showing evaluator-specific tier bias. **c**, Normalized prediction entropy of the full predicted distribution, where higher values indicate broader use of the four tiers and lower values indicate stronger collapse into a narrow subset of labels.



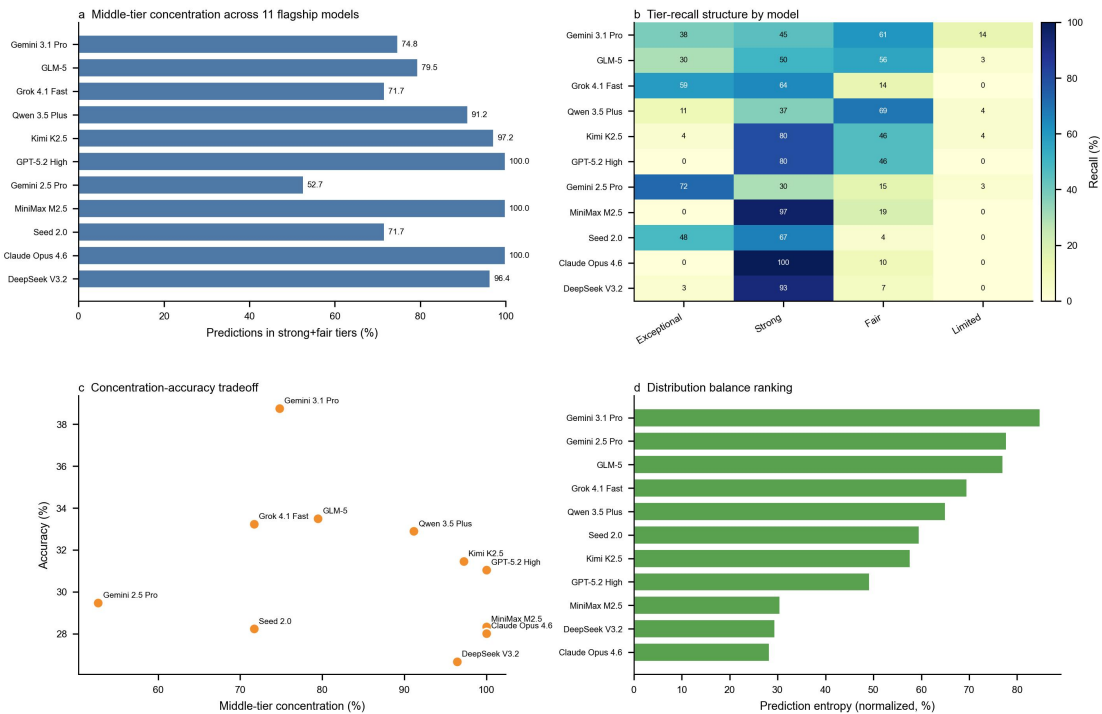
Supplementary Figure 2 (SF2): Expert Individual Accuracy Distribution

a, Histogram of per-expert accuracy for the unfiltered expert panel ($N = 48$; 8 pitches each). Dashed line indicates mean accuracy (36.2%); dotted line indicates chance level (25%). The panel highlights substantial heterogeneity across experts, not a tight cluster around a common level of skill. **b**, Empirical cumulative distribution function (CDF) of per-expert accuracy with the same chance baseline, making it easier to see how much of the expert panel lies near chance versus in the higher-performing tail.



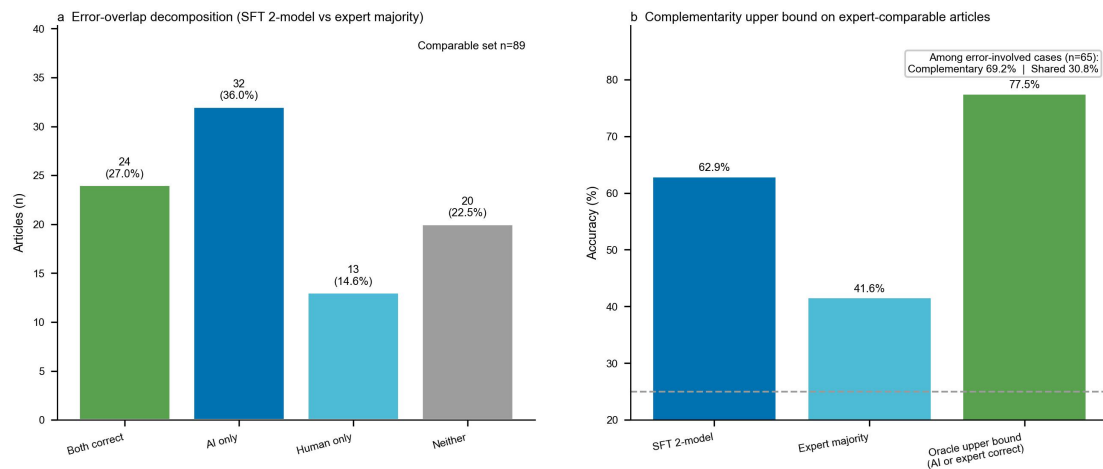
Supplementary Figure 3 (SF3): Junior Monte Carlo Subsampling Curve

a, Majority-vote accuracy versus panel size under repeated Monte Carlo subsampling (5,000 draws), with 95% confidence band. Chance baseline (25%) is shown as a dotted line, and the curve shows that larger panels help early but then flatten. **b**, Marginal accuracy gain per additional rater, showing diminishing returns as panel size increases and clarifying why aggregation alone does not keep improving linearly.



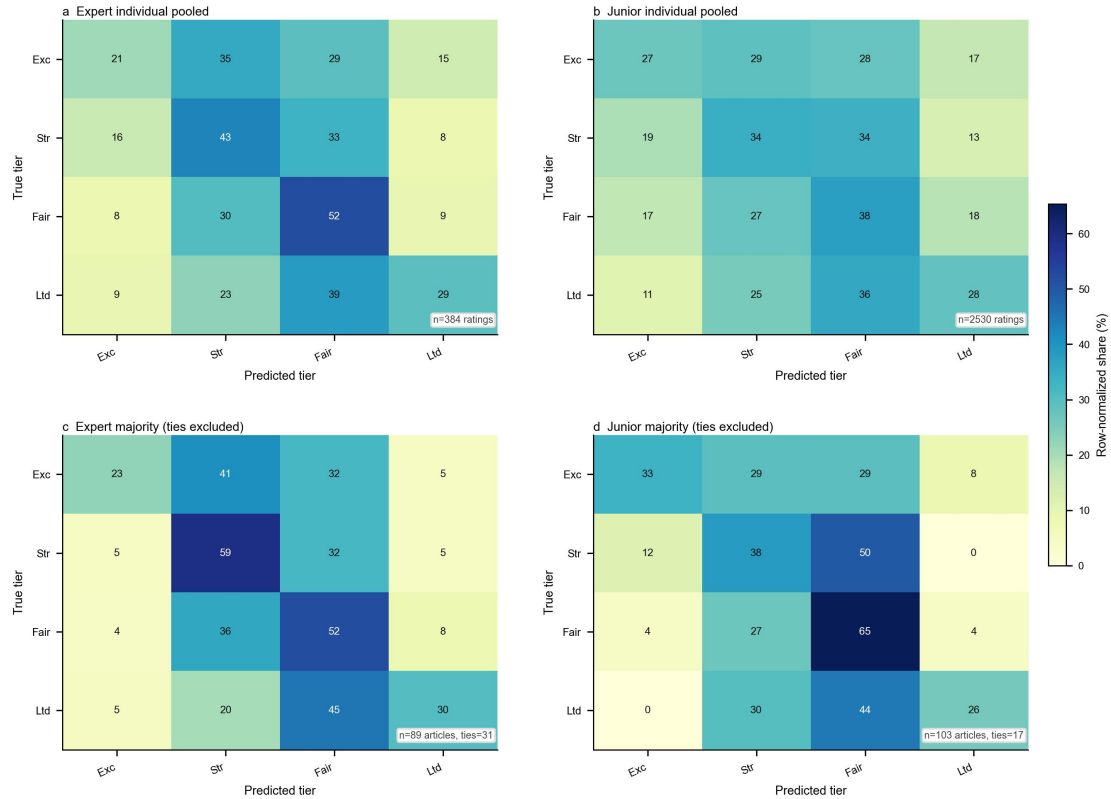
Supplementary Figure 4 (SF4): Frontier Collapse-Metric Landscape

Collapse diagnostics for the 11 frontier models. **a**, Share of predictions assigned to the middle tiers (strong + fair) by model. **b**, Per-tier recall heatmap across the four quality tiers, making visible which classes are effectively never recovered. **c**, Relationship between middle-tier concentration and overall accuracy across models. **d**, Normalized prediction entropy ranking, where higher values indicate less distributional collapse and broader use of the four-tier scale.



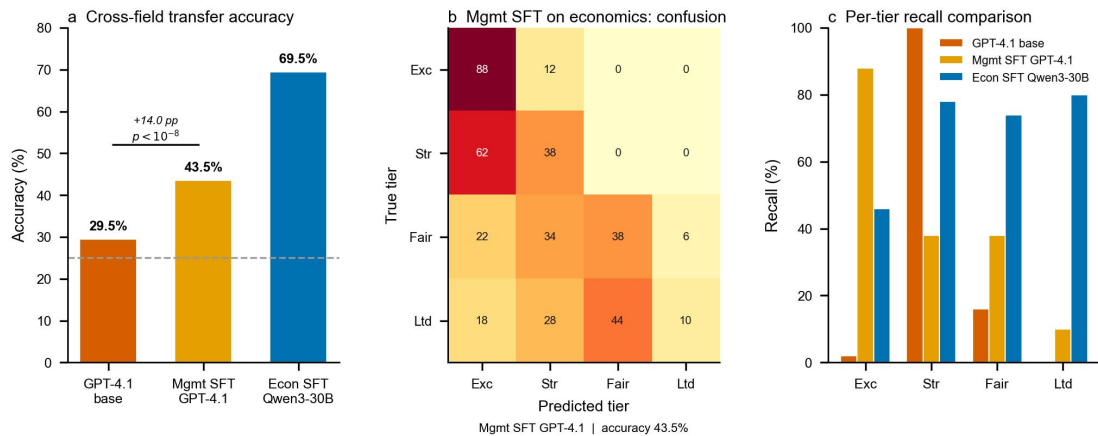
Supplementary Figure 5 (SF5): AI-Human Error Complementarity

a, Overlap decomposition of correct and incorrect outcomes between the SFT ensemble and expert majority vote on the expert-comparable subset ($N = 89$ non-tied pitches): both correct, AI-only correct, human-only correct, and shared error. This panel shows directly how much of the two systems' success is overlapping versus complementary. **b**, Complementarity ceiling: the oracle upper bound reached when either AI or the expert majority is correct is 77.5%, compared with SFT alone (62.9% on this subset) and expert majority vote (41.6%), quantifying the remaining room for hybrid routing strategies.



Supplementary Figure 6 (SF6): Human Confusion Matrices

Row-normalized confusion matrices for expert and junior panels. **a**, Expert individual pooled ($N = 384$ ratings). **b**, Junior individual pooled ($N = 2,530$ ratings). **c**, Expert strict clear-majority voting ($N = 89$ non-tied; 31 ties excluded). **d**, Junior strict clear-majority voting ($N = 103$ non-tied; 17 ties excluded). Reading pooled and majority panels together clarifies how much disagreement is smoothed by voting and which off-diagonal confusions persist even after aggregation. Panel composition is summarized in Supplementary Table ST5, and majority-vote counts are summarized in Supplementary Table ST7.



Supplementary Figure 7 (SF7): Cross-Field Transfer from Management to Economics

The management-trained SFT GPT-4.1 checkpoint (trained exclusively on management institutional traces) is evaluated on the 200-article economics benchmark without any economics-specific fine-tuning. **a**, Overall accuracy comparison: GPT-4.1 base (29.5%, not significantly above chance), management-trained SFT GPT-4.1 (43.5%, $p < 10^{-8}$ versus chance; +14.0 percentage points over base), and the best economics in-domain SFT (Qwen3-30B, 69.5%) for reference. **b**, Row-normalized confusion matrix for the management SFT on economics, showing 88% exceptional recall but only 10% limited recall, consistent with

management quality signals transferring best at the top of the quality distribution. **c**, Per-tier recall comparison across all three evaluators, showing that the management SFT captures partial cross-field signal concentrated in the upper tiers, while the base model collapses into middle-tier predictions.

Supplementary References

Supplementary citations use the same numbered bibliography as the main manuscript.